

Linear Regression

$$Y = mX + c$$

Grate Smashers (Youtube)

Pizza:

$$Y = mX + b$$

Diameter (x) In Inches	price (y) In Dollars	Mean (x)	Mean (y)	Deviations (x)	Deviations (y)	product of Deviations	Sum of product of Deviations	Square of Deviations for X
8	10	10	13	$8-10 = -2$	$10-13 = -3$	$-2 \times -3 = 6$	12	$(-2)^2 = 4$
10	13	10	13	$10-10 = 0$	$13-13 = 0$	$0 \times 0 = 0$	0	0
12	16	10	13	$12-10 = 2$	$16-13 = 3$	$2 \times 3 = 6$	12	4

Calculate $m = \text{Sum of product of deviations} / \text{Sum of square of deviation for } x$

$$= 12/8$$

$$= 1.5$$

→ अर्थात् 'x' पर value 1 बाढ़ाने 'y' पर मान 1.5 घूबे इन्दि रहे ।

Calculate $b = \text{Mean of } Y - (m * \text{Mean of } X)$

=

$$Y = mX + b$$

$$= 13 - (1.5 \times 10)$$

$$= 13 - 15$$

$$= -2$$

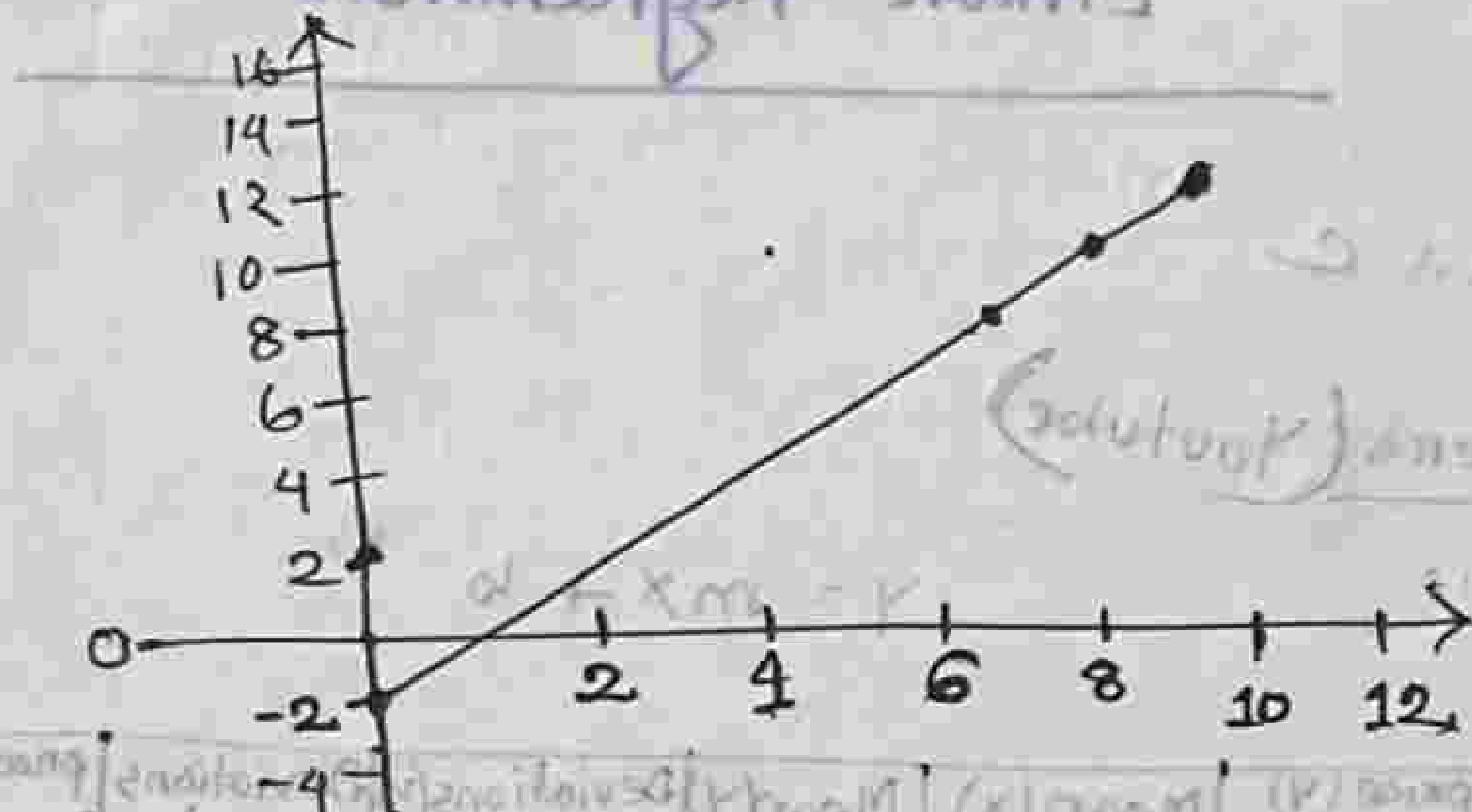
$$\rightarrow X = 1$$

X पर

value 0 रहे

Y पर value -2

Linear Regression

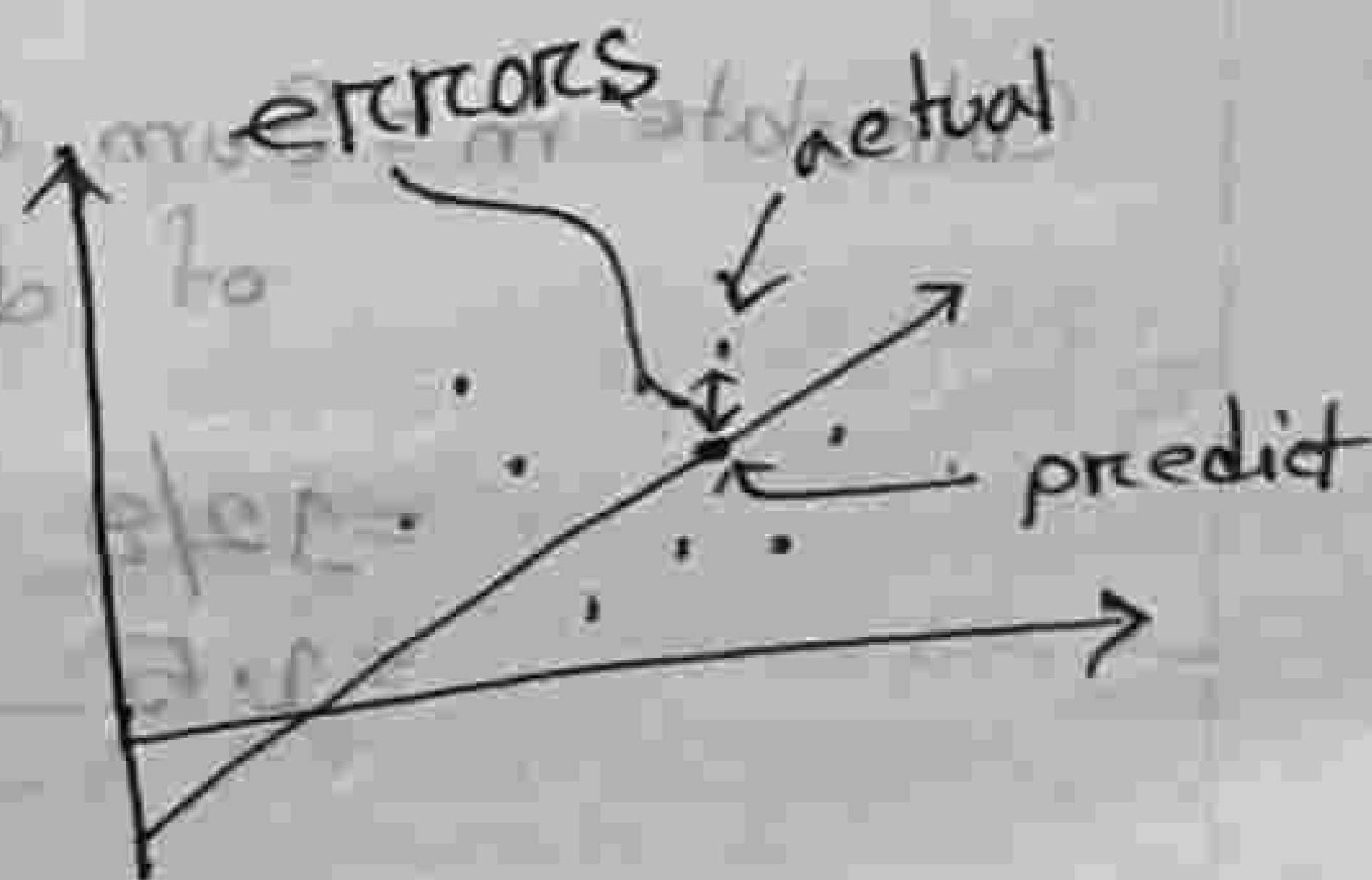


Let, assume diameter 20 inches. Now predict the price.

$$y = 1.5 \times 20 - 2$$

$$= 30 - 2$$

$$= 28$$



$$\text{Calculate } b = \frac{\text{Mean of } y - (\text{mean of } x) \times m}{\text{Mean of } x^2 - (\text{mean of } x)^2}$$

$$b = \frac{13 - (1.5 \times 10)}{13 - 10}$$

$$= \frac{13 - 15}{13 - 10}$$

$$= \frac{-2}{3}$$

like value -5
value 0 error
x do

Independent value $\rightarrow$ continuous
dependent $\rightarrow$ discrete

Sigmoid:

$$Y = \frac{1}{1 + e^{-(a_0 + a_1 x)}}$$

From slide 2:

$$MSE = \frac{1}{2n} \sum (\hat{y}_i - y_i)^2$$

কতটা ভুল হচ্ছে তা
মাপতে পারে।

$$\hat{y}_i = mx + c$$

$y_i \rightarrow$ actual
value.

[Cost কম $\rightarrow$ model ভালো]

example from chatgpt:

Output (দাম) continuous value হলে (সটে Regression.

Simple Linear Regression:

Given, $m = 0.05$, $c = 10$, $x = 1000$

$$\hat{y} = mx + c$$

$$\hat{y} = ?$$

$$\hat{y} = 0.05 \times 1000 + 10 = 60$$

Cost Function (MSE):

$$\text{Actual} = [50, 60], \text{Predicted} = [48, 63]$$

$$\text{MSE} = \frac{(50-48)^2 + (60-63)^2}{2 \times \textcircled{2}} = 6.5 \quad 3.25$$

$\downarrow$
 n

~~2n & n ko kr same krdo. kr n~~
~~3.25 krdo.~~

$$= 3.25$$

Gradient Descent:

$$\text{Given, } m = 0.5, \alpha = 0.01, \text{gradient}_m = 4$$

$$\text{undated } m = ?$$

$$m_{\text{new}} = 0.5 \times -0.01 \times 4 = 0.46$$

$$\text{if, } \boxed{m - \alpha \frac{\partial J}{\partial m}} \rightarrow \text{gradient}_m$$

$$\text{gradient}_c = -3.5 \quad \text{then } C_{\text{new}} = ?$$

$$C_{\text{new}} = 0.5 + 0.01 \times 3.5$$
$$= 2.035$$

$$m - \alpha \frac{\partial J}{\partial c}$$

$\downarrow$
gradient_c

Vectorized Linear Regression:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, W = \begin{bmatrix} c \\ m \end{bmatrix}$$

$$\hat{y} = XW \rightarrow \text{predict value.}$$

Question:

$$X = \begin{bmatrix} 1 & 100 \\ 1 & 200 \end{bmatrix}, W = \begin{bmatrix} 10 \\ 0.5 \end{bmatrix}$$

predict value = ?

Soln:

$$\hat{y} = XW = \begin{bmatrix} 60 \\ 110 \end{bmatrix}$$

Multiple Linear Regression:

Question:

$$\hat{y} = 10 + 0.04x_1 + 5x_2$$

$$x_1 = 1000, x_2 = 3, \hat{y} \text{ is?}$$

Soln:

$$\hat{y} = 10 + 40 + 15 = 65$$

polynomial Regression:

Q. Given: $\hat{y} = 2x^2 + 3x + 5$
 $x = 2$; $\hat{y} = ?$

$$\hat{y} = 2(4) + 3 \times 2 + 5 = 19$$

* Over & Under fitting!
↓
সুন্দর
করেছে previous

Overfitting:

Model training data খুব ভালো ফিট করে,
কিন্তু নতুন (test) data-তে খারাপ করে।

Ridge Regression (L2 Regularization):

Cost function এ L2 penalty (w^2) যোগ করে
weight ছোট করে। এতে model জটিলতা
হয় এবং Overfitting করে।

$$J = \text{MSE} + \lambda \sum w^2$$

ex:

উদাহরণ:

$$w_1 = 15, w_2 = 10$$

Ridge পরে: $w_1 = 4, w_2 = 3$

Lasso Regression (L1 Regularization):

cost-এ L1 penalty ($|w|$) যোগ করে কিছু feature বাদ দেয়।

$$J = \text{MSE} - \lambda \sum |w|$$

ex:

উদাহরণ:

$$w_1 = 5, w_2 = 2, w_3 = 1$$

পরে:

$$w_1 = 3, w_2 = 0, w_3 = 0$$

[Feature 2 & 3 বাদ]

Lasso vs Ridge:

	Ridge (L2)	Lasso (L1)
penalty	w^2	$ w $
weight	ছোট হয়	সহজ শূন্য হয়ে
Feature selection	না	হ্যাঁ
example	অথ feature shrink	কিছু feature বাদ

Real ex:

Features = age, income, zip_code

Ridge $\rightarrow$ সব রাখে

Lasso $\rightarrow$ zip_code বাদ দিতে পারে

কিন্তু লস (loss) $(\|w\|_1)$ কমানোর জন্য

$$J = MSE - \lambda \|w\|_1$$

$$J = MSE, \lambda = 0, \lambda = 100$$

$$J = MSE, \lambda = 0, \lambda = 100$$

[Feature selection]

Lasso vs Ridge

(a) Lasso

(b) Ridge

অন্য ফিচার

শুধু

feature

weight

feature selection

example

feature selection

অন্য ফিচার

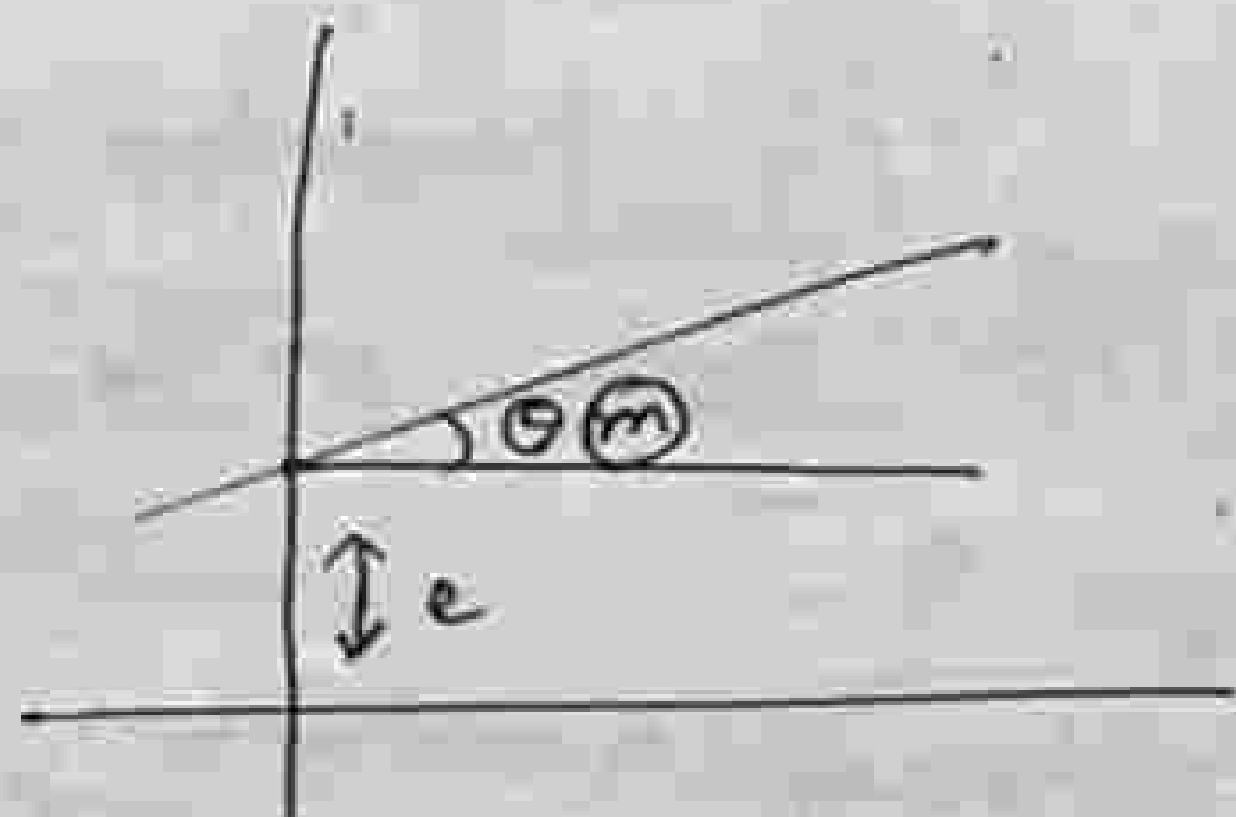
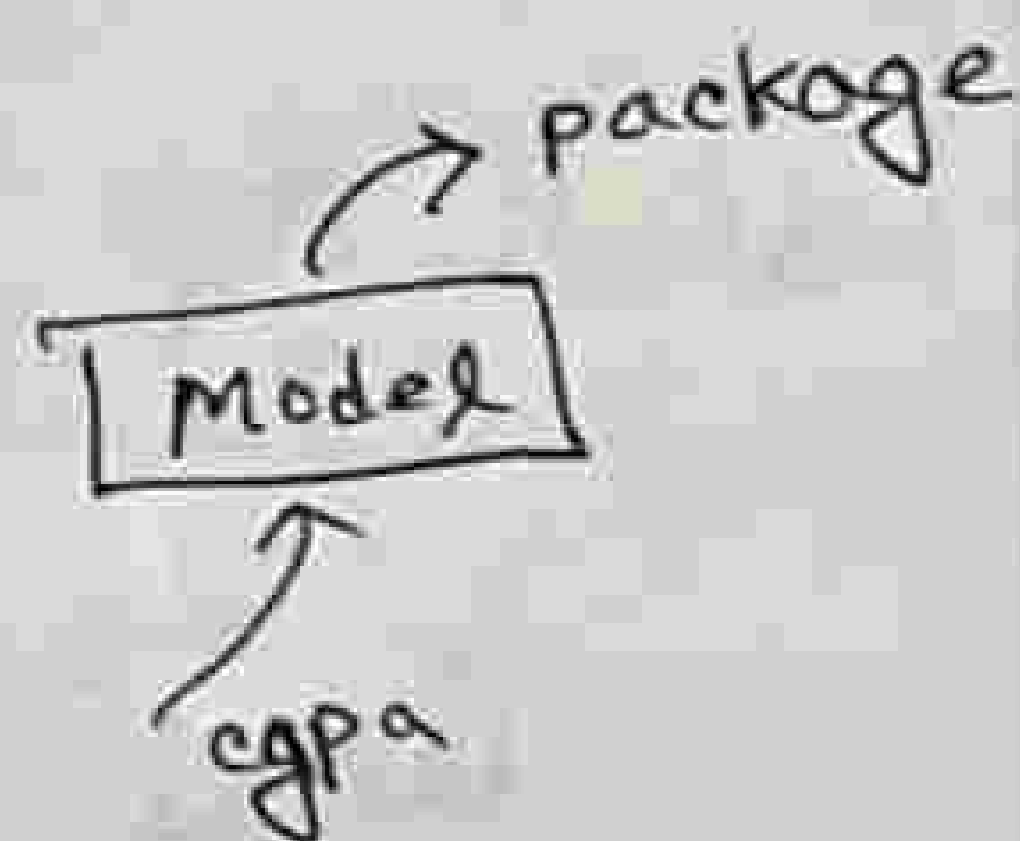
→ CampusX

Simple Linear Regression:

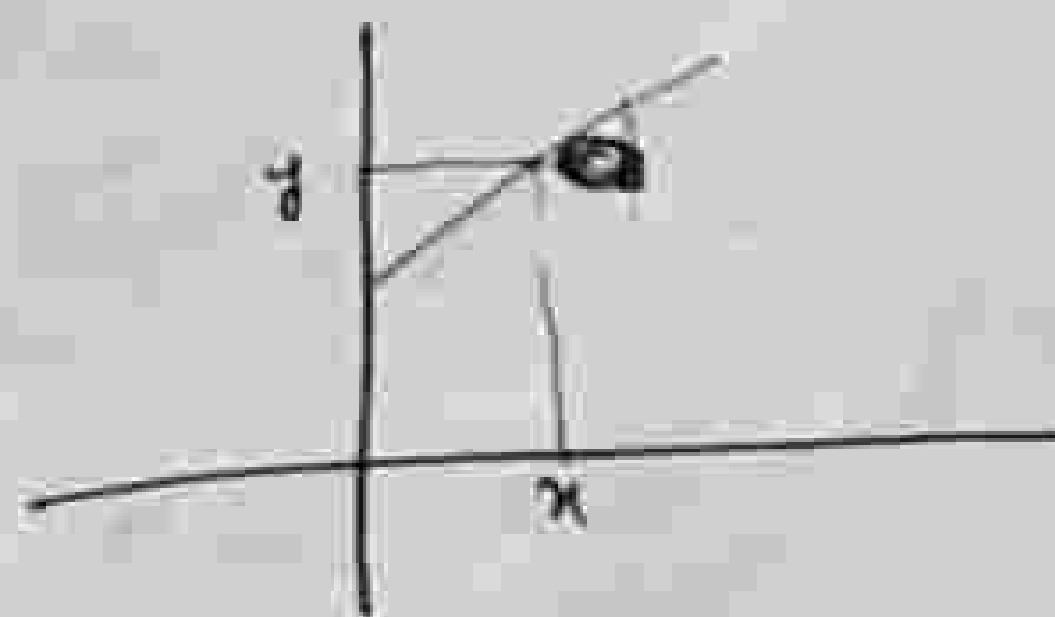
1 input column | 1 output column

ex:

cgpa	package
7.1	3.5
4.7	1.2
8.9	4.2
8.1	3.9



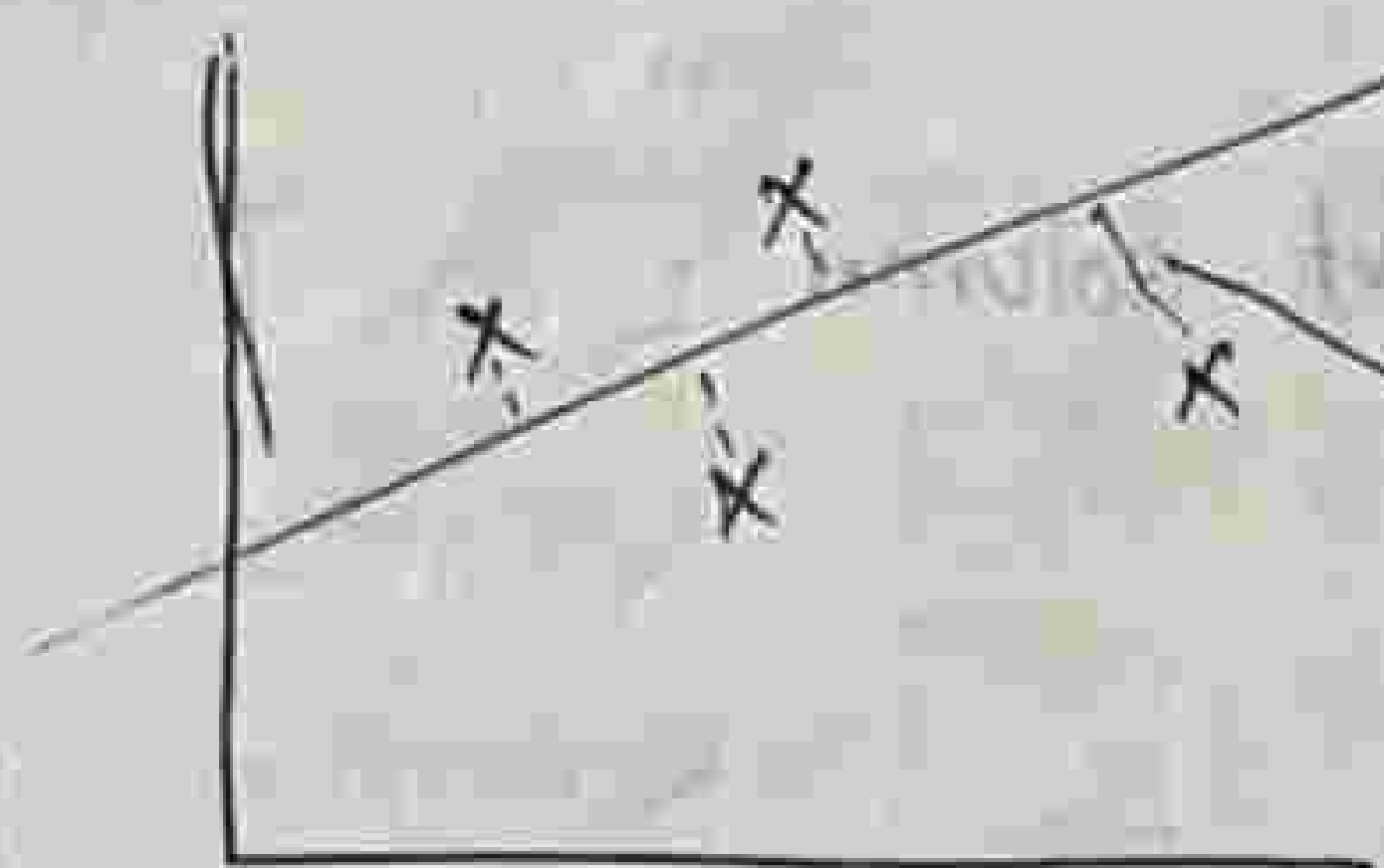
$$y = mx + e$$



→ sort of data line draw

Carboxy

perfect line draw कर।



कार distance को minimise कर।

प्रयोग

प्रयोग

प्रयोग

प्रयोग	प्रयोग
2.8	1.8
5.1	5.1
0.1	0.1
0.8	1.8

$$y + x \cdot m = b$$



प्रयोग प्रयोग प्रयोग

* Decision Tree ::

AI ৯ম থেকে

steps to follow

① choose a target Attribute.

② calculate I.G. (Information gain) of that target attribute.

③ Calculate Entropy of other attributes using the following formula,

$$E(A) = \sum_{i=1}^N \frac{p_i + N_i}{p + N} \times I(p_i, N_i)$$

④ Subtract $E(A)$ from I.G. of each attribute for find out the Gain.

$$\text{Entropy} = \frac{-p}{p+N} \log_2 \left(\frac{p}{p+N} \right) - \frac{N}{p+N} \log_2 \left(\frac{N}{p+N} \right)$$

$$\text{Gain} = \text{I.G.} - E(A)$$

Ex:-

target - खिला

Weather	Humidity	Wind	play
Sunny	High	Weak	No
Sunny	Normal	Weak	No
Sunny	Normal	Strong	No
Cloudy	High	Weak	No
Cloudy	High	Strong	No
Cloudy	Normal	Strong	Yes
Cloudy	Normal	Weak	Yes
Rainy	High	Weak	Yes
Rainy	Normal	Strong	Yes
Rainy	Normal	Weak	Yes

Soln:

Let,

Target attributes = play

Information ^{Entropy} Gain for play:

$$I_{\text{Entropy}} = - \frac{P}{P+N} \log_2 \left(\frac{P}{P+N} \right) - \frac{N}{P+N} \log_2 \left(\frac{N}{P+N} \right)$$

$$= - \frac{5}{5+5} \log_2 \left(\frac{5}{5+5} \right) - \frac{5}{5+5} \log_2 \frac{5}{5+5}$$

$$= - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right)$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

Yes/No का
अंतर (5+5)
N/P का
अंतर 1

→ concept is clear
but calculation is not clear

Calculating Entropy,

For weather,

	No	Yes
Sunny	3	0
Cloudy	2	2
Rainy	0	3

I. G. ko probability
Entropy ko probability
probability ko value
dina hai.

$$\begin{aligned} \text{Entropy}_{\text{I.G.}}(\text{Sunny}) &= -\frac{3}{3+0} \log_2\left(\frac{3}{3+0}\right) - \frac{0}{3} \log_2\left(\frac{0}{3}\right) \\ &= 0 \times \frac{3}{10} = 0 \text{ Entropy} \end{aligned}$$

→ I. G. ko sunny ko probability diana
dina hai entropy

$$\text{Entropy}_{\text{I.G.}}(\text{cloudy}) = -\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right)$$

$$= 1$$

$$\text{Entropy} = 1 \times \frac{4}{10} = 0.4$$

→ 4 = cloudy ka count

10 = total instances

$$\begin{aligned} \text{Entropy}_{\text{I.G.}}(\text{Rainy}) &= -\frac{0}{3} \log_2\left(\frac{0}{3}\right) - \frac{3}{3} \log_2\left(\frac{3}{3}\right) \\ &= 0 \end{aligned}$$

$$\text{Entropy} = 0 \times \frac{3}{10} = 0$$

$$E(\text{weather}) = 0 + 0.4 + 0 = 0.4$$

$$(\text{weather}) \text{ Gain} = 1 - 0.4 = 0.6$$

$$\text{Gain}(\text{weather}) = 1 - \left[0 \times \frac{3}{10} + 1 \times \frac{4}{10} + 0 \times \frac{3}{10}\right] = 0.6$$

For Humidity,

	No	Yes
High	3	1
Normal	2	4

$$\text{Entropy}_{\text{Total}}(\text{High}) = -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right)$$

$$= -\frac{3}{4} (-0.415) - \frac{1}{4} (-2) = 0.813$$

$$E(\text{High}) = 0.813 \times \frac{4}{10} = 0.325$$

$$E(\text{Normal}) = -\frac{2}{6} \log_2 \left(\frac{2}{6} \right) - \frac{4}{6} \log_2 \left(\frac{4}{6} \right)$$

$$= -\frac{1}{3} (-1.585) - \frac{2}{3} (-0.585)$$

$$= 0.9183$$

$$E(\text{Normal}) = 0.9183 \times \frac{6}{10} = 0.551$$

$$E(\text{Humidity}) = 0.325 + 0.551 = 0.876$$

$$(\text{Humidity}) \text{ Gain} = 1 - 0.876 = 0.124$$

→ Total Entropy

For Wind,

	No	Yes	
Weak	3	3	= 6
Strong	2	2	= 4

$$I.G.(\text{weak}) = -\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) = 1$$

$$E(\text{weak}) = 1 \times \frac{6}{10} = 0.6$$

$$I.G.(\text{strong}) = -2/4 \log_2(2/4) - 2/4 \log_2(2/4) = 1$$

$$E(\text{strong}) = 1 \times \frac{4}{10} = 0.4$$

$$E(\text{Wind}) = 0.6 + 0.4 = 1$$

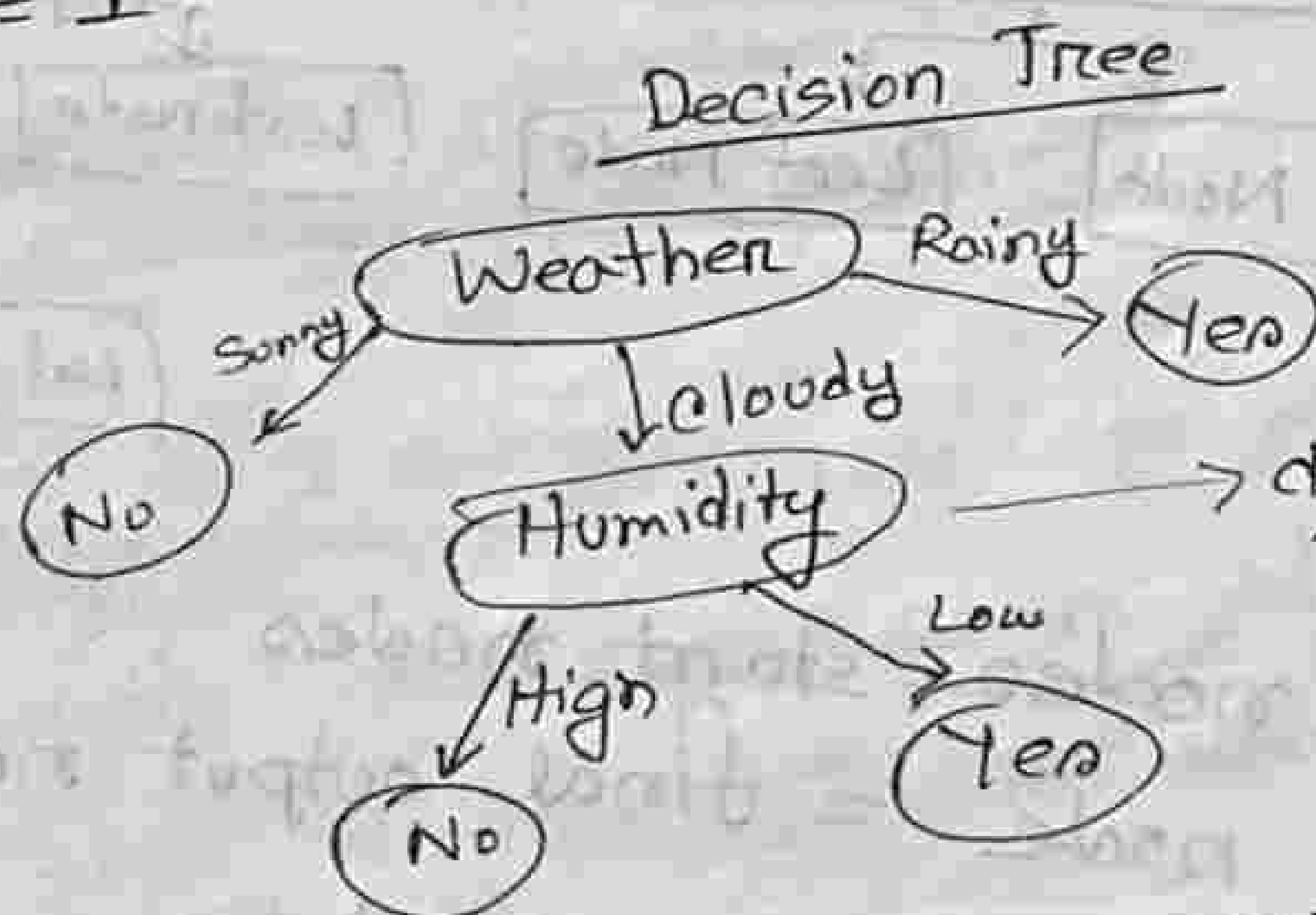
$$(\text{Wind}) \text{ Gain} = 1 - 1 = 0$$

$$\text{Now, Gain}(\text{Weather}) = 0.6$$

$$\text{Gain}(\text{Humidity}) = 0.124$$

$$\text{Gain}(\text{Wind}) = 0$$

$$I.G. = 1$$



(Weather is the root node because it has the highest gain)

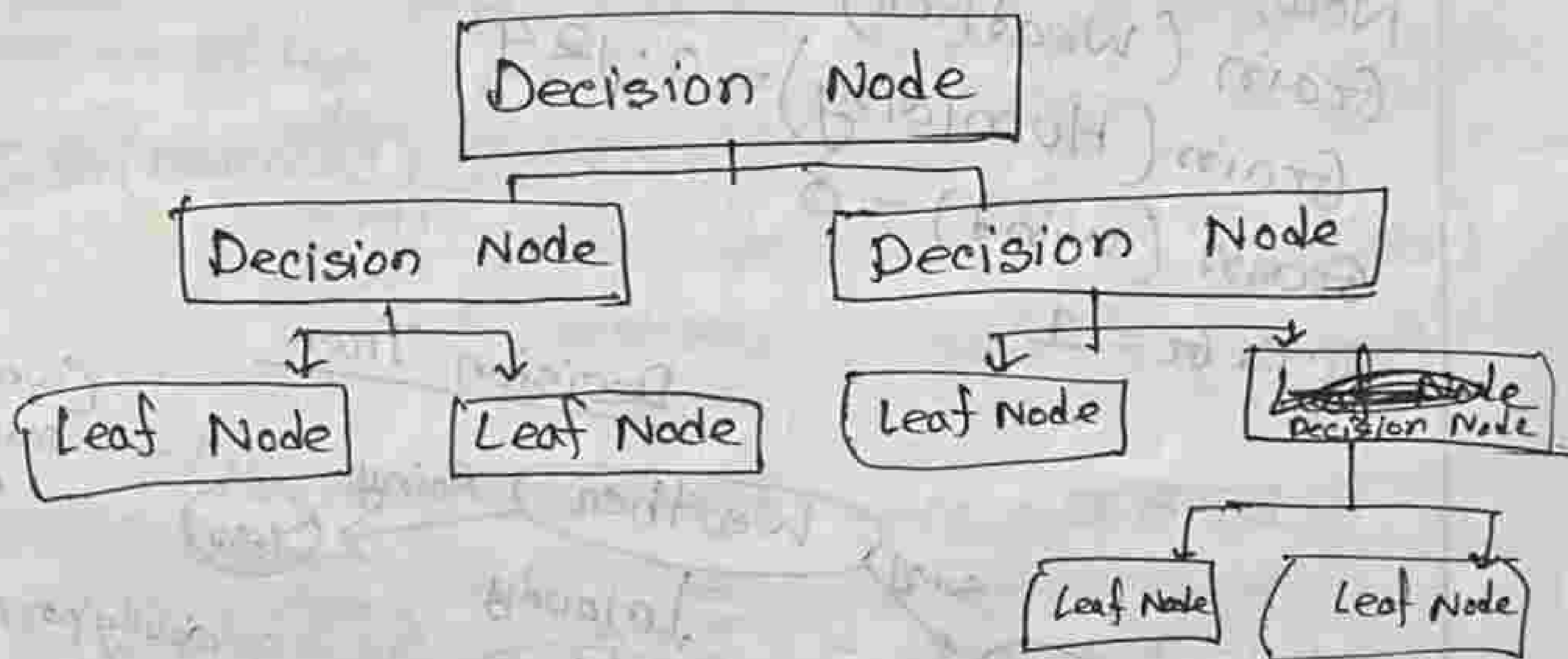
cloudy is No/Yes mix so we split on Humidity to get the best gain

Humidity is not the best split because it has a low gain.

* Introduction of Decision Tree:

- is a supervised learning technique.
- Internal nodes represent the features of a dataset. Branches represent the decision rules, and external nodes represents the outcome.

Terminologies:



Root nodes = start nodes

Leaf node = final output node

Decision Node = used to make any decision and have multiple branches.

Branch/sub tree = splitting the tree.

Pruning = removing the unwanted branches from the tree.

parent/child node:

root = parent & others are child node

* Attributes Selection Measures (ASM)

I. G

(Information gain)

Entropy

→ how to select the best attribute for the root node and for sub-nodes.

* Entropy: is a metric to measure the impurity in a given attribute. It is the degree of randomness in data.

(Randomness
ताकिता सता
high)
entropy

Low entropy

Low entropy

high entropy

* I.G: Measurement of changes in entropy after the segmentation of a dataset based on an attribute.

example from slide:

Target attributes = play Tennis

Entropy Gain for Play Tennis:

$$I.G = - \frac{P}{P+N} \log_2 \left(\frac{P}{P+N} \right) - \frac{N}{P+N} \log_2 \left(\frac{N}{P+N} \right)$$

$$= - \frac{9}{9+5} \log_2 \left(\frac{9}{14} \right) - \frac{5}{9+5} \log_2 \left(\frac{5}{14} \right)$$

$$= - \frac{9}{14} \log_2 = - \frac{9}{14} \times (-0.19189) - \frac{5}{14} \times (-0.4472)$$

$$= 0.123358 + 0.1597$$

$$= 0.283058$$

$$= - \frac{9}{14} \times (-0.63743) - \frac{5}{14} \times (-1.48543)$$

$$= 0.4098$$

$$+ 0.53$$

$$= 0.9498$$

Calculating Entropy for each attributes:

For Outlook,

	No	Yes
Sunny	3	2
Overcast	0	4
Rain	3	2

$$E_{Info}(Sunny) = - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right)$$

$$= 0.52877 + 0.4422$$

$$= 0.970$$

$$E(Sunny) = 0.970 \times \frac{5}{14} = 0.347$$

$$E_{Info}(Overcast) = - \frac{4}{4+0} \log_2 \left(\frac{4}{4} \right) - \frac{0}{4+0} \log_2 \left(\frac{0}{4} \right)$$

$$= 0 - 0 = 0$$

$$E(Overcast) = 0 \times 4/14 = 0$$

$$E_{Info}(Rain) = - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right)$$

$$= 0.5288 + 0.4422$$

$$= 0.970$$

$$E(Rain) = 0.970 \times \frac{5}{14} = 0.347$$

$$E(Outlook) = 0.347 + 0 + 0.347$$

$$= 0.694$$

$$\text{Outlook (Gain)} = 0.9498 - 0.694$$

$$= 0.2558$$

For, Temperature,

	Yes	No
hot	2	2
cool	3	1
mild	4	2

$$I.G.(hot) = -\frac{2}{4} \log_2(2/4) - \frac{2}{4} \log_2(2/4)$$

$$= 0.5 + 0.5 = 1$$

$$E(hot) = 1 \times \frac{4}{14} = 0.28572$$

$$I.G.(cool) = -\frac{3}{4} \log_2(3/4) - \frac{1}{4} \log_2(1/4)$$

$$= 0.3113 + 0.5$$

$$= 0.8113$$

$$E(cool) = 0.8113 \times \frac{4}{14} = 0.2318$$

$$I.G.(mild) = -\frac{4}{6} \log_2(4/6) - \frac{2}{6} \log_2(2/6)$$

$$= 0.38997 + 0.52832$$

$$= 0.91829$$

$$E(mild) = 0.39355$$

$$E(\text{Temperature}) = 0.911$$

$$\text{Temperature (brain)} = 0.9498 - 0.911$$

$$= 0.039$$

For humidity,

	Yes	No
high	3	4
Normal	6	1

$$I.G.(high) = -\frac{3}{7} \log_2(3/7) - \frac{4}{7} \log_2(4/7)$$

$$= 0.5239 + 0.46135$$

$$= 0.98525$$

$$E(high) = 0.4926$$

$$I.G.(\text{Normal}) = -\frac{6}{7} \log_2(6/7) - \frac{1}{7} \log_2(1/7)$$

$$= 0.190 + 0.4$$

$$= 0.591$$

$$E(\text{Normal}) = 0.29553$$

$$E(\text{Humidity}) = 0.788$$

$$\text{Humidity (gain)} = 0.9498 - 0.788 = 0.16167$$

For Wind,

	Yes	No
Weak	5	2
Strong	3	3
False	1	0

$$I.G.(weak) = -\frac{5}{7} \log_2(5/7) - \frac{2}{7} \log_2(2/7)$$

$$= 0.3467 + 0.5164$$

$$= 0.863$$

$$E(Weak) = 0.863 \times \frac{7}{14} = 0.43157$$

Entropy
probability
কম (less)
বিস্তারিত

$$I.G. (strong) = -\frac{3}{6} \log_2(3/6) - \frac{3}{6} \log_2(3/6) \\ = 0.5 + 0.5 = 1$$

$$E(strong) = 1 \times \frac{6}{14} = 0.4286$$

$$I.G. (False) = -\frac{1}{1+0} \log_2(1) - \frac{0}{1} \log_2(0/1) \\ = 0$$

$$E(wind) = 0.43157 + 0.4286 + 0 \\ = 0.86017$$

$$Wind (Gain) = 0.9498 - 0.86017 \\ = 0.08963$$

Now,

$$Outlook Gain = 0.2558$$

$$Temperature Gain = 0.039$$

$$Humidity Gain = 0.16167$$

$$Wind Gain = 0.08963$$

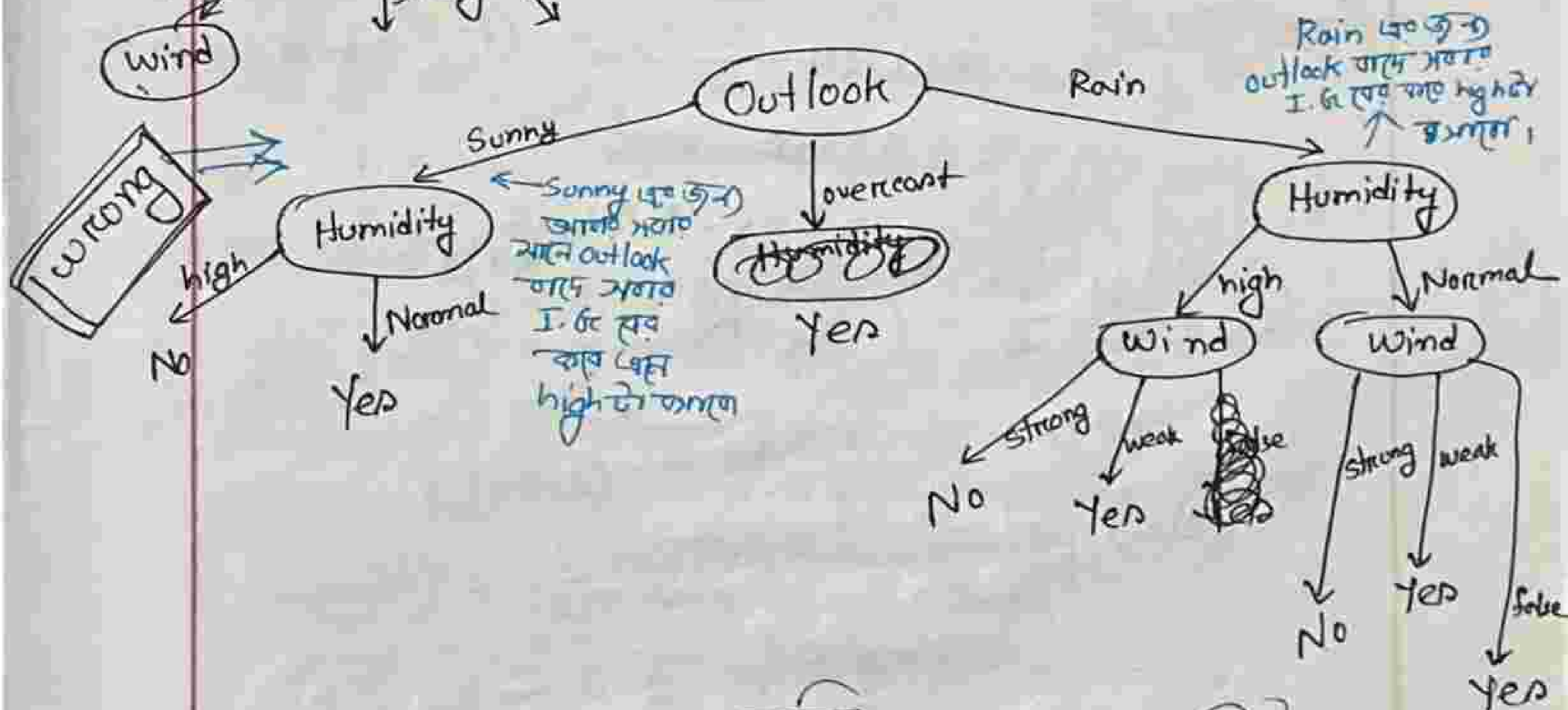
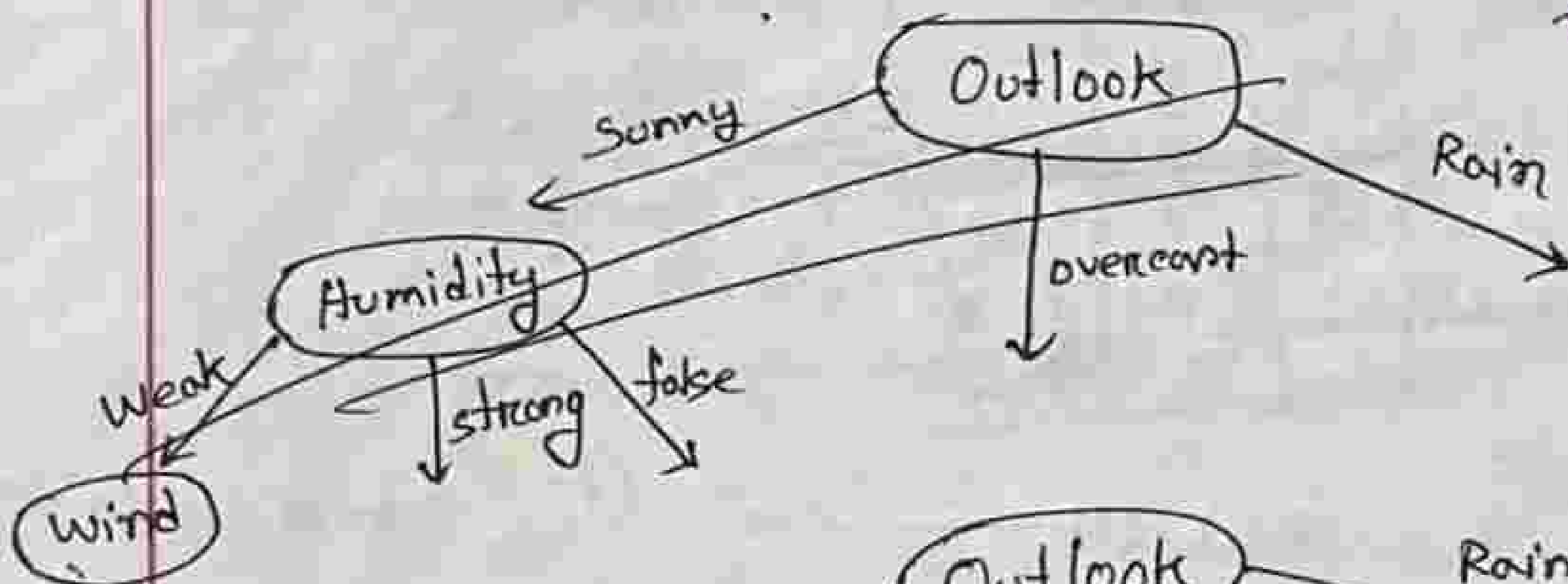
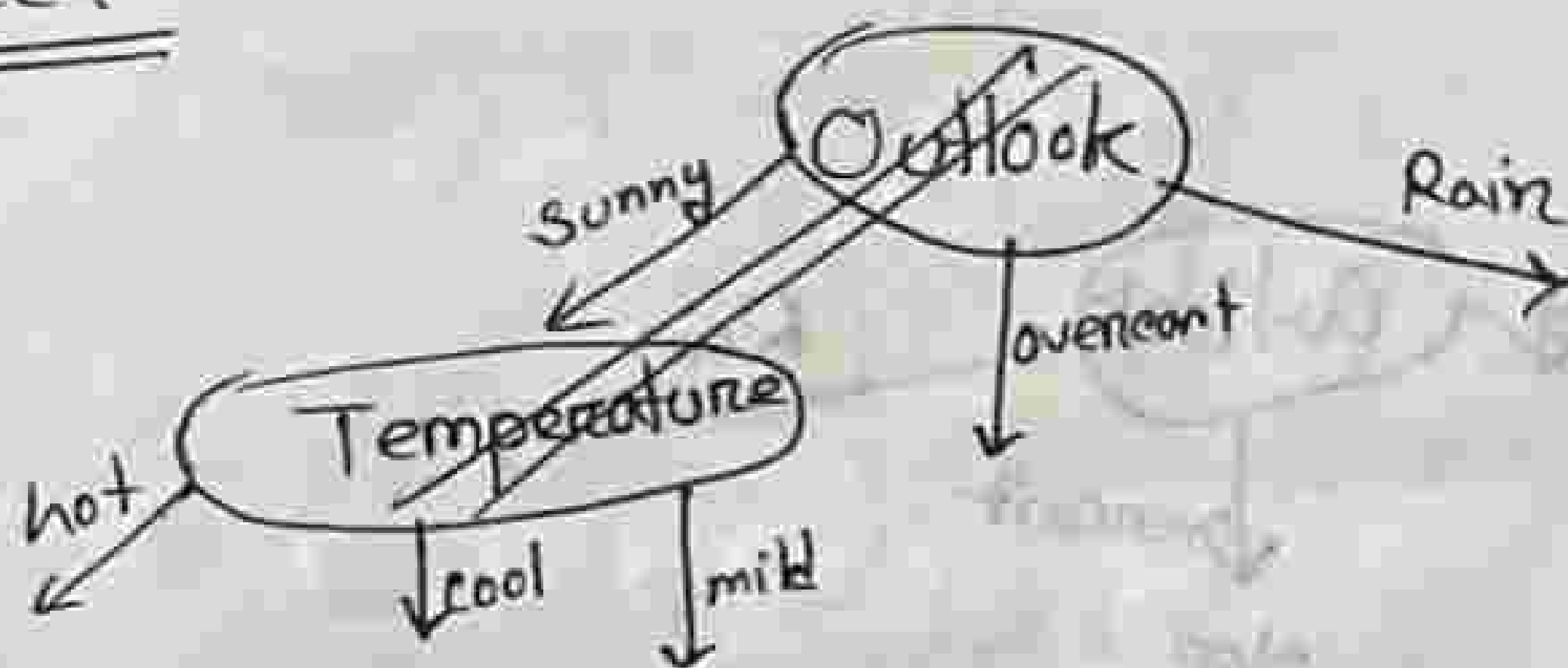
	Yes	No	Strong
Outlook	2	2	2
Temperature	2	2	2
Humidity	2	2	2
Wind	2	2	2

(P/E)

2/13 = 0.1538

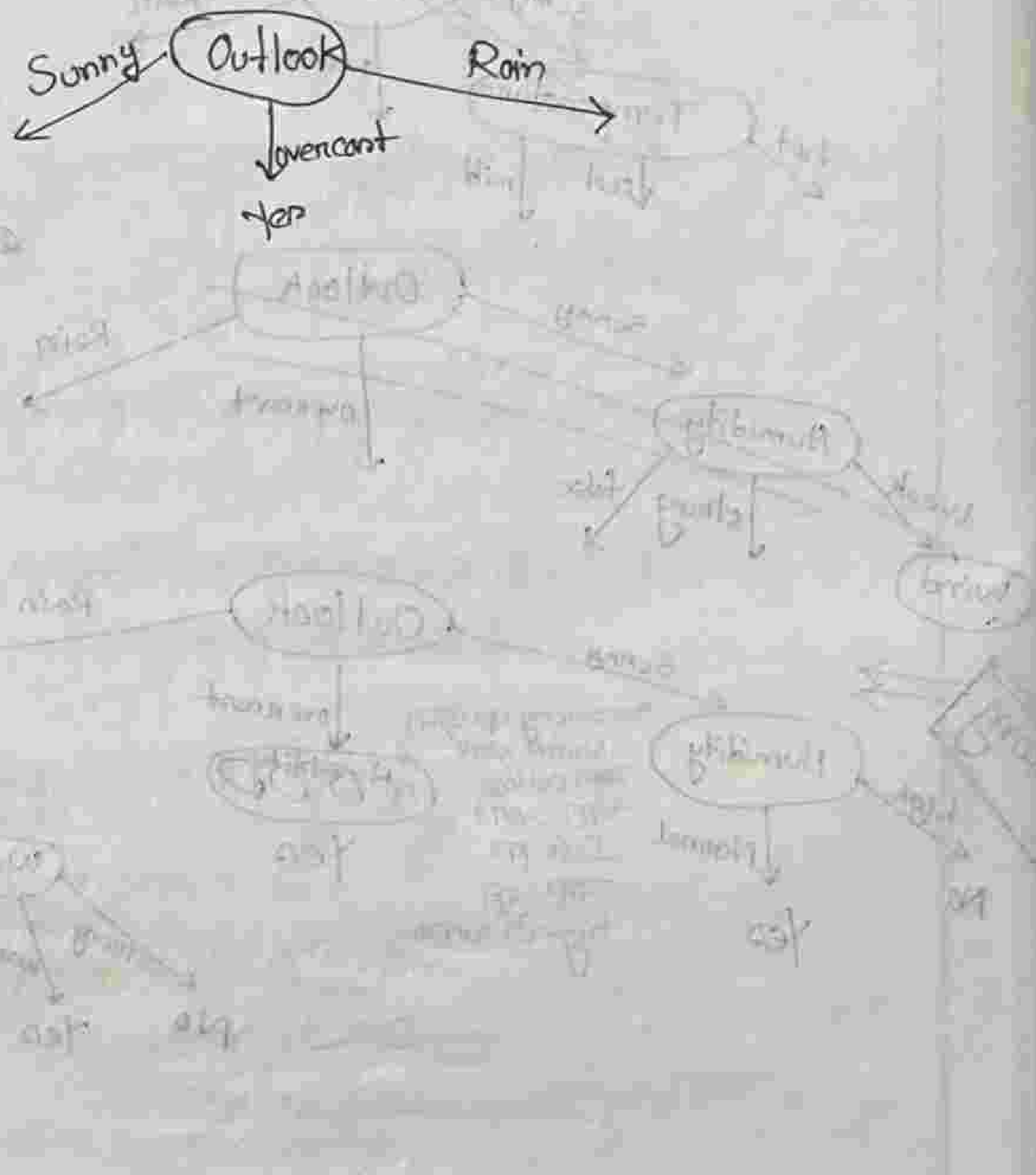
$$E(wind) = \frac{P}{N} \times 2.585 = 0.1538 \times 2.585 = 0.396$$

Tree:



જાણીએ શું ની । હવાન પ્રાણીને ધાલ
આવા I. 6. calculate જાણે રા

Decision tree:



Decision tree is a model for classification and regression analysis. It is a flowchart-like structure, where each internal node represents a "test" on an attribute (e.g., numerical value or name), each branch represents an outcome of the test, and each leaf node represents a class assignment or numerical value.

Entropy

Calculation:

For sunny,

	Yes	No
	2	3

For Temperature,

	Yes	No
hot	0	2
cool	1	0
mild	1	1

$$I_{\text{bit}}(\text{hot}) = -\frac{0}{2} \log_2(0/2) - \frac{2}{2} \log_2(2/2)$$

$$= 0 - 0$$

$$E(\text{hot}) = 0$$

$$I_{\text{bit}}(\text{cool}) = -\frac{1}{1} \log_2(1) - \frac{0}{1} \log_2(0/1)$$

$$E(\text{cool}) = 0$$

$$I_{\text{bit}}(\text{mild}) = -\frac{1}{2} \log_2(1/2) - \frac{1}{2} \log_2(1/2)$$

$$= 0.5 + 0.5 = 1$$

$$E(\text{mild}) = 1 \times \frac{2}{5} = 0.4$$

sunny
सunny

$$E(\text{Temperature}) = 0.4$$

$$\text{Temperature (Gain)} = 0.9498 - 0.4 = 0.5498$$

For Humidity,

	Yes	No
high	0	3
Normal	2	0

$$I.G. (high) = -\frac{3}{5} \log_2(3/5) - 0$$

$$= 0.4422$$

$$I.G. (high) = 0.4422 \times \frac{3}{5} = 0.2653$$

$$I.G. (Normal) = -\frac{2}{2} \log_2(2/2) - 0$$

$$= 0$$

$$E(Normal) = 0$$

$$E(Humidity) = 0.2653$$

$$Humidity (gain) = 0.9498 - 0.2653$$

$$= 0.6845$$

For Wind,

⌀

	Yes	No
Weak	1	2
Strong	1	1
False	1	0

$$I.G. (Weak) = -\frac{1}{3} \log_2(1/3) - \frac{2}{3} \log_2(2/3)$$

$$= 0.52832 + 0.389975$$

$$= 0.9183$$

$$E(Weak) = 0.9183 \times \frac{3}{5} = 0.55$$

$$I.G. (strong) = 1$$

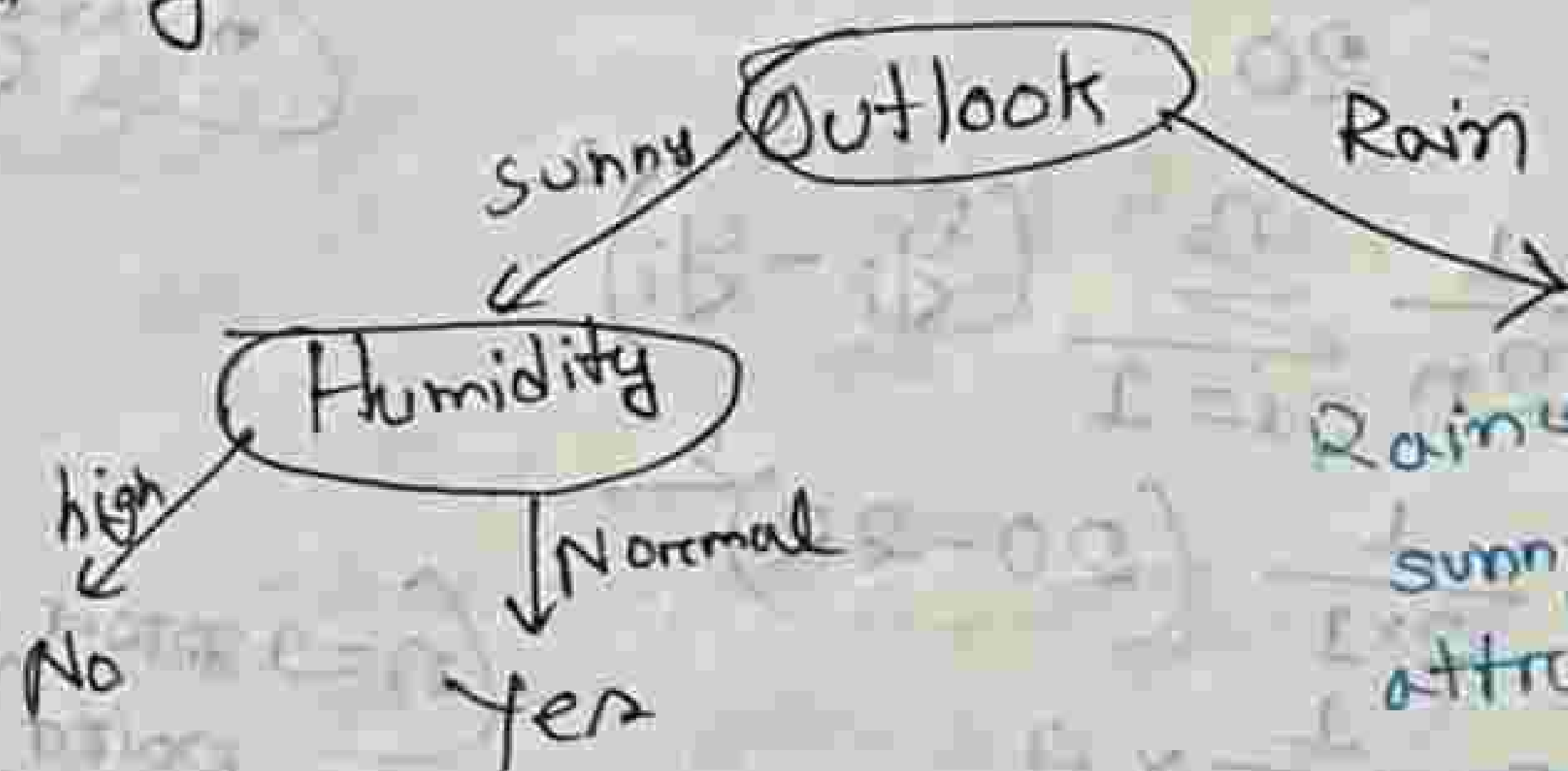
$$I.G. (false) E(strong) = 1 \times \frac{2}{5} = 0.4$$

$$I.G. (false) = 0$$

$$I.G. (Wind) = 0.55$$

$$I.G. (Rain) = 0.9498 - 0.55 = 0.3998$$

আমরা Outlook থেকে Sunny এর জন্য বাকি attributes
 খুঁজতে থাকি (Temperature, Humidity, Wind) এর মধ্যে
 Humidity এর Rain বাকি, তাই tree



Linear Regression slide:

ex: Given a model with $m=2.5$ and $c=10$: Predict the value of $\{y\}$ when $x=4$. If the actual value y for $x=4$ is 22, calculate the squared error for this single point.

Solⁿ:

$$\hat{y} = mx + c$$

$$= 2.5 \times 4 + 10$$

$$= 20$$

$\hat{y}$ = predicted value.

m = weight/slope

$$MSE = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

$$= \frac{1}{2 \times 1} (20 - 22)^2$$

$$= \frac{1}{2} \times 4$$

$$= 2 \text{ (Ans)}$$

$n=1$ कारण single point

AI

ex: You are building a model to predict house prices with 50 different features, but you suspect only 5 or 6 of them actually matter. Which regularization technique (Ridge or Lasso) should you use, and why?

* Vectorized Form of Linear Regression (Batch Gradient Descent):

Solⁿ: Recommendation: Lasso Regression.

Because it will force the coefficients of the 44-45 "useless" features to become exactly zero.

$$\left[\begin{matrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{matrix} \right] \frac{1}{2 \times 50} = \frac{1}{100}$$

(higher values)

ex: Given dataset,

i	x	y
1	1	2
2	2	3
3	3	4

$x = \text{study hours}$
 $y = \text{cgpa}$

$$\hat{y} = mx + c$$

Initialize $m=0, c=0$.

prediction ($\hat{y}$):

x	y	$\hat{y}$
1	2	0
2	3	0
3	4	0

$n=3$

Cost function, $J = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$

↙
MSE

$$= \frac{1}{2 \times 3} [(0-2)^2 + (0-3)^2 + (0-4)^2]$$

$$= 4.83 \text{ (higher errors)}$$

Compute Gradients,

$$\begin{aligned}\text{Slope (m)} &= \frac{\partial J}{\partial m} = \frac{1}{n} \sum (\hat{y}_i - y_i) x_i \\ &= \frac{1}{3} [(0-2) \times 1 + (0-3) \times 2 + (0-4) \times 3] \\ &= \frac{1}{3} [-2 - 6 - 12] = \frac{1}{3} \times (-20) \\ &= -6.67\end{aligned}$$

$$\begin{aligned}\text{Intercept (c)} &= \frac{\partial J}{\partial c} = \frac{1}{n} \sum (\hat{y}_i - y_i) \\ &= \frac{1}{3} \times [(0-2) + (0-3) + (0-4)] \\ &= \frac{1}{3} \times (-9) = -3\end{aligned}$$

Gradient Descent update,

$$m := m - \alpha \frac{\partial J}{\partial m} \quad \text{Let, } \alpha = 0.1$$

$$\begin{aligned}&= 0 - (0.1) \times (-6.67) \\ &= 0.67\end{aligned}$$

$$c := c - \alpha \frac{\partial J}{\partial c}$$

$$\begin{aligned}&= 0 - (0.1) \times (-3) \\ &= 0.3\end{aligned}$$

New prediction after iteration,

$$\hat{y}_i = 0.67x + 0.3$$

x	y	$\hat{y}$
1	2	1.93
2	3	3.26
3	4	4.59

→ $\hat{y}_i$ କରାଯାଇ
ଏখানে calculate କରିବି
demo ହେବି ବିନା

Just process the
ମିଳାଯାଇଥିବା
previous eqⁿ (ଅଳ)
right to the end
ମିଳି ଥାଏ।

Now, cont,

$$J = \frac{1}{2n} \sum (y_i - \hat{y}_i)^2$$

$$= \frac{1}{2 \times 3} [(1.93 - 2)^2 + (3.26 - 3)^2 + (4.59 - 4)^2]$$

$$= 0.07$$

Ex: Linear Regression with vectorized

Dataset:

sample	Units(x)	Bill(y)
1	1	5
2	2	9
3	3	13

here, $n=3$

① Design matrix, $\rightarrow C$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 9 \\ 13 \end{bmatrix}$$

initialize,

$$w = \begin{bmatrix} c \\ m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

vectorized

prediction, $\hat{y} = Xw$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$Xw - y = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 5 \\ 9 \\ 13 \end{bmatrix}$$

$$= \begin{bmatrix} -5 \\ -9 \\ -13 \end{bmatrix}$$

Ex: Linear Regression with vectorized

Dataset:

sample	Units(x)	Bill(y)
1	1	5
2	2	9
3	3	13

here, $n=3$

i) Design matrix, $\rightarrow c$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 9 \\ 13 \end{bmatrix}$$

initialize, $w = \begin{bmatrix} c \\ m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

vectorized prediction, $\hat{y} = Xw$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now, $Xw - y = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 5 \\ 9 \\ 13 \end{bmatrix}$

$$= \begin{bmatrix} -5 \\ -9 \\ -13 \end{bmatrix}$$

$$\text{MSE, } J(w) = \frac{1}{2n} (Xw - y)^T (Xw - y)$$

$$= \frac{1}{6} \times \begin{bmatrix} -5 & -9 & -13 \end{bmatrix} \times \begin{bmatrix} -5 \\ -9 \\ -13 \end{bmatrix}$$

$$= \frac{1}{6} \times \begin{bmatrix} 25 + 45 + 65 \\ 5^2 + 9^2 + 13^2 \end{bmatrix}$$

$$= \frac{1}{6} \times 275$$

$$= 45.833$$

Vectorized Gradient,

$$\nabla J(w) = \frac{1}{n} X^T (Xw - y)$$

$$= \frac{1}{3} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \times \begin{bmatrix} -5 \\ -9 \\ -13 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} -5 - 9 - 13 \\ -5 - 18 - 39 \end{bmatrix}$$

$$= \begin{bmatrix} -9 \\ -20.67 \end{bmatrix}$$

Gradient Descent Update

$$w := w - \alpha \nabla J(w)$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.01 \times \begin{bmatrix} -9 \\ -20.67 \end{bmatrix}$$

$$= \begin{bmatrix} 0.09 \\ 0.2067 \end{bmatrix}$$

$$(\alpha = 0.01)$$

New Prediction (after one iteration),

$$\begin{aligned}\hat{y} &= Xu \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0.09 \\ 0.2067 \end{bmatrix} \\ &= \begin{bmatrix} 0.2967 \\ 0.3034 \\ 0.7101 \end{bmatrix}\end{aligned}$$

(continued)

Multiple Linear Regression:

$$\hat{y}_i = w_0 + w_1 f_{i1} + w_2 f_{i2} + \dots + w_n f_{in}$$

i	x_1	x_2	y
1	800	2	150
2	1000	3	200
3	1200	4	250

Here,

$$n=3$$

$$d=2$$

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2$$

Now,

$$X = \begin{bmatrix} 1 & 800 & 2 \\ 1 & 1000 & 3 \\ 1 & 1200 & 4 \end{bmatrix}; \quad y = \begin{bmatrix} 150 \\ 200 \\ 250 \end{bmatrix}$$

initialize weight,

$$w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

prediction,

$$\hat{y} = Xw$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Vectorized Cost Function,

$$J(w) = \frac{1}{2n} (Xw - y)^T (Xw - y)$$

$$= \frac{1}{2 \cdot 3} [(-150)^2 + (-200)^2 + (-250)^2]$$

$$= 20833.33$$

Vectorized Gradient,

$$\Delta J(w) = \frac{1}{n} X^T (Xw - y)$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 800 & 1000 & 1200 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} -150 \\ -200 \\ -250 \end{bmatrix}$$

$$= \begin{bmatrix} -200 \\ -2066.67 \\ -566.67 \end{bmatrix}$$

Gradient descent update,

$$w := w - \alpha J(w)$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -200 \\ -206666.67 \\ -566.67 \end{bmatrix} \times 0.001$$

$$= \begin{bmatrix} 2 \\ 2066.67 \\ 567 \end{bmatrix}$$

New Prediction,

$$\hat{y} = Xw = \begin{bmatrix} 1 & 800 & 2 \\ 1 & 1000 & 3 \\ 1 & 1200 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 2066.67 \\ 567 \end{bmatrix}$$

All equation for vectorized (method):

$$\textcircled{1} X = \begin{bmatrix} 1 & f_{11} \\ 1 & f_{21} \\ \vdots & \vdots \\ 1 & f_{n1} \end{bmatrix}$$

$$w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}$$

$$\hat{y} = Xw \rightarrow \text{prediction}$$

Cost function,

$$J(w) = \frac{1}{2n} (Xw - y)^T (Xw - y)$$

Vectorized Gradient $\Delta w J = \frac{1}{n} X^T (Xw - y)$

Gradient Descent Update,

$$w := w - \alpha \Delta w J$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 20 \\ 26 \end{bmatrix}$$

All equations for vectorized (matrix):

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 20 \\ 26 \end{bmatrix} \quad \text{①}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 20 \\ 26 \end{bmatrix}$$

$$\text{output} \leftarrow \text{net} = \beta$$

All formula from Regression slide

Simple Linear Reg:

$$\hat{y} = mx + c \rightarrow \text{predict value}$$

$$\text{MSE} \div J = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \rightarrow \text{cost function}$$

Gradient,

$$\frac{\partial J}{\partial m} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i$$

or,

$$\frac{\partial J}{\partial c} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \rightarrow \text{intercept}$$

Update Gradient & intercept;

$$m := m - \alpha \frac{\partial J}{\partial m} ; c := c - \alpha \frac{\partial J}{\partial c}$$

Vectorized Form of Linear Regression

$$X = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} ; \omega = \begin{bmatrix} c \\ m \end{bmatrix} ;$$

predict value, $\hat{y} = X\omega$

$$\text{Cost Function, } J(\omega) = \frac{1}{2n} (X\omega - y)^T (X\omega - y)$$

$$\text{Gradients, } \nabla_{\omega} J = \frac{1}{n} X^T (X\omega - y)$$

Gradient Descent Update,

$$\omega := \omega - \alpha \nabla_{\omega} J$$

Multiple Linear Regression:

$$\hat{y}_i = w_0 + w_1 f_{i1} + w_2 f_{i2} + \dots + w_d f_{id}$$

Matrix Feature,

$$X = \begin{bmatrix} 1 & f_{11} & f_{12} & \dots & f_{1d} \\ 1 & f_{21} & f_{22} & \dots & f_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & f_{n1} & f_{n2} & \dots & f_{nd} \end{bmatrix}$$

$$w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix}$$

prediction, $\hat{y} = Xw$

$$\text{Cost Function, } J(w) = \frac{1}{2n} (Xw - y)^T (Xw - y)$$

$$\text{Gradient, } \nabla_w J(w) = \frac{1}{n} X^T (Xw - y)$$

Gradient Descent Update, :

$$w := w - \eta \nabla_w J(w)$$

$$(y - wx)^T (y - wx) \frac{1}{2n} = J(w)$$

$$(y - wx)^T x \frac{1}{n} = \nabla J(w)$$

$$w := w - \eta \nabla J(w) = w - \eta \frac{1}{n} X^T (Xw - y)$$

Logistic Regression

→ 0-1 (binary) class

prediction (sigmoid function), $\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$

$$z = xw$$

Cost Function,

$$L(w) = - \left[y \log(\hat{y}) + (1-y) \log(1-\hat{y}) \right]$$

↪ single

$$L(w) = - \frac{1}{n} \sum_{i=1}^n \left[y^{(i)} \log(\hat{y}^{(i)}) + (1-y^{(i)}) \log(1-\hat{y}^{(i)}) \right]$$

Gradient,

$$\frac{\partial J(w)}{\partial w} = \frac{1}{n} \sum_{i=1}^n \left(\hat{y}^{(i)} - y^{(i)} \right) f^{(i)}$$

update gradient,

$$w' = w - \alpha \nabla_w J$$

$$z = xw$$

ex: $x = [2, 3]$, $w = [0.1, -0.2]$, $b = 0$

$\hat{y} = ?$

$\bar{z} = xw = [2, 3] \times [0.1, -0.2]$
 $= (2 \times 0.1) + (3 \times (-0.2))$

$= -0.4$

$\hat{y} = \frac{1}{1 + e^{-\bar{z}}} = 0.401 \approx 40.1\%$

$w := w - \alpha \nabla_w J$

ex: Given, $x = \begin{bmatrix} 1 & 5 \\ 2 & 1 \\ 3 & 3 \end{bmatrix}$, $w = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}$

$y = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$\alpha = 0.1$

$w_{\text{new}} = ?$

Soln:

$\bar{z} = xw = \begin{bmatrix} 1 & 5 \\ 2 & 1 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}$
 $= \begin{bmatrix} 0.2 + 0.5 \\ 0.4 + 0.1 \\ 0.6 + 0.3 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.5 \\ 0.9 \end{bmatrix}$

$$\hat{y} = \begin{bmatrix} \frac{1}{1+e^{-0.7}} \\ \frac{1}{1+e^{-0.5}} \\ \frac{1}{1+e^{-0.9}} \end{bmatrix} = \begin{bmatrix} 0.67 \\ 0.62 \\ 0.71 \end{bmatrix}$$

$$\text{Error vector} = \begin{bmatrix} 0.67 - 1 \\ 0.62 - 0 \\ 0.71 - 1 \end{bmatrix} = \begin{bmatrix} -0.33 \\ 0.62 \\ -0.29 \end{bmatrix}$$

$$\begin{aligned} \nabla_w J &= \frac{1}{n} X^T (\hat{y} - y) \\ &= \frac{1}{3} \begin{bmatrix} 1 & 2 & 3 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} -0.33 \\ 0.62 \\ -0.29 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{3} \begin{bmatrix} (1 \times -0.33) + (2 \times 0.62) + (3 \times -0.29) \\ (5 \times -0.33) + (1 \times 0.62) + (3 \times -0.29) \end{bmatrix} \\ &= \begin{bmatrix} 0.013 \\ 0.1633 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} w_{\text{new}} &= w - \alpha \nabla_w J = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix} - 0.1 \begin{bmatrix} 0.013 \\ 0.1633 \end{bmatrix} \\ &= \begin{bmatrix} 0.1987 \\ 0.1633 \end{bmatrix} \end{aligned}$$

10-12-2025

Practice Example:

Linear Regression

f_1	f_2	y
2	3	7
3	4	8
5	6	9

dataset

$$w = \begin{bmatrix} -1 \\ 0.2 \\ 0.3 \end{bmatrix}$$

$$\alpha = 0.01$$

after 5 iteration

prediction.

Soln:

$$\hat{y} = Xw$$

$$\nabla_w J = \frac{1}{n} X^T (Xw - y)$$

$$w = w - \alpha \frac{1}{n} X^T (Xw - y)$$

it-1:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 5 & 6 \end{bmatrix}$$

$$\hat{y} = Xw = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 5 & 6 \end{bmatrix} \begin{bmatrix} -1 \\ 0.2 \\ 0.3 \end{bmatrix}$$

$$\begin{bmatrix} -1 + 2 \times 0.2 + 3 \times 0.3 \\ -1 + 3 \times 0.2 + 4 \times 0.3 \\ -1 + 5 \times 0.2 + 6 \times 0.3 \end{bmatrix}$$

$$= \begin{bmatrix} 0.3 \\ 0.2 \\ 1.8 \end{bmatrix}$$

$$\nabla_{\omega} J = \frac{1}{n} X^T (\hat{y} - y)$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 5 \\ 3 & 4 & 6 \end{bmatrix} \left(\begin{bmatrix} 0.3 \\ 0.2 \\ 1.8 \end{bmatrix} - \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \right)$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 5 \\ 3 & 4 & 6 \end{bmatrix} \begin{bmatrix} -6.7 \\ -7.2 \\ -7.2 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} -21.1 \\ -7.1 \\ -9.2 \end{bmatrix}$$

$$\omega = \omega - \alpha \nabla_{\omega} J$$

$$= \begin{bmatrix} -1 \\ 0.2 \\ 0.3 \end{bmatrix} - \begin{bmatrix} -0.07 \\ -0.025 \\ -0.03 \end{bmatrix}$$

$$= \begin{bmatrix} -0.9 \\ 0.223 \\ 0.303 \end{bmatrix}$$

$$\begin{bmatrix} 6.0 \times 8 + 5.0 \times 5 + 1.0 \\ 8.0 \times 12 + 5.0 \times 8 + 1.0 \\ 8.0 \times 12 + 6.0 \times 8 + 1.0 \end{bmatrix}$$

05-01-2025

DT $\rightarrow$ Decision Tree

pure class = ১ম feature class

Impure = ~~এ~~ মিশ্র থাকলে।

$$\text{Entropy} = \sum_i -p_j \log_2 p_j$$

$$\text{Gini Index} = 1 - \sum_i p_j^2$$

$$\text{Classification error} = 1 - \max\{p_j\}$$

2.2 Geometric Interpretation:

$$p = \frac{y(x)}{|w|}$$

→ boundary এর distance বোঝানো যায়,

$$p = \frac{|w^T x + w_0|}{\sqrt{w_1^2 + w_2^2 + \dots + w_n^2}}$$

→ আনকগুলো হলো

Ex-2.1:

Find the distance from point

$x = (3, 4)^T$ to the decision boundary

defined by $2x_1 + x_2 - 5 = 0$

Soln:

$$y(x) = w^T x + w_0 \quad \left| \begin{array}{l} w=? \\ w_0=? \end{array} \right.$$

here,

$$w_0 = -5$$

$$w = (2, 1)^T$$

↳ coefficient

Now,

~~$$\gamma(x) = (2, 1)^T \cdot (3, 4)^T - 5$$~~

$$\gamma(x) = (2, 1)^T \cdot (3, 4)^T - 5$$

vector আকারে লিখলে vector এর সূত্র হবে।

~~$$= (2 \times 3 + 1 \times 4) - 5$$~~

$$= (6 + 4) - 5 = 10 - 5$$

$$= 5$$

$\therefore \text{class} = C_1 (\because > 0)$

$$D = \frac{\gamma(x)}{|W|} = \frac{5}{\sqrt{4+1}} = \frac{5}{\sqrt{5}} = \sqrt{5} \text{ (Ans.)}$$

Ex-2.2: Consider the decision boundary $x_1 - 2x_2 + 3 = 0$. Find the perpendicular distance from the following points and classify them:

Ⓐ $x_a = (1, 1)^T$

Ⓑ $x_b = (-1, 2)^T$

Ⓒ $x_c = (3, 3)^T$

Soln:

For $x_a = (1, 1)^T$:

$$x_1 - 2x_2 + 3 = 0$$

$$w = (1, -2)^T$$

$$w_0 = 3$$

$$\gamma(x) = w^T x + w_0$$

$$(1, -2)^T \cdot (1, 1)^T + 3$$

$$= (1 - 2) + 3$$

$$= 2$$

$$h = \frac{2}{\sqrt{1+4}}$$

$$x_0 = (1, 1)^T$$

$$x_1 = (2, 1)^T$$

$$x_2 = (3, 2)^T$$

For $x_b = (-1, 2)^T$:

$$\gamma(x) = w^T x + w_0$$

$$= (1, -2)^T (-1, 2)^T + 3$$

$$= (-1 - 4) + 3$$

$$= -5 + 3$$

$$= -2$$

$$= -2$$

$$b = \frac{-2}{\sqrt{1+4}} = \frac{-2}{\sqrt{5}}$$

For $x_c = (3, 3)^T$:

$$\gamma(x) = w^T x + w_0$$

$$= (1, -2)^T (3, 3)^T + 3$$

$$= (3 - 6) + 3$$

$$= -3 + 3$$

$$= 0$$

$$= 0$$

$$b = 0$$

→ exactly on the decision surface.

Ex-2.3 = How

Two parallel decision boundaries are defined by $2x_1 + x_2 - 4 = 0$ and $2x_1 + x_2 + 2 = 0$

Find the perpendicular distance between two boundaries.

Soln:

page 2: derivation

Consider a point x and its projection $x_{\perp}$ onto the decision boundary. The vector from $x_{\perp}$ to x is parallel to w :

$$x = x_{\perp} + b \frac{w}{\|w\|}$$

Since $x_{\perp}$ is on the boundary, $w^T x_{\perp} + w_0 = 0$. Substituting

$$\begin{aligned} y(x) &= w^T x + w_0 = w^T \left(x_{\perp} + b \frac{w}{\|w\|} \right) + w_0 \\ &= w^T x_{\perp} + w_0 + b \frac{w^T w}{\|w\|} = 0 + b \frac{\|w\|^2}{\|w\|} = b \|w\| \end{aligned}$$

$\downarrow$
 $x_{\perp}$ is on the boundary.

Therefore,

$$y(x) = b \|w\|$$

$$b = \frac{y(x)}{\|w\|}$$

recorded
video

2/2 marks

mid-2:

10-4-2020 (online class)

Ensemble Learning:

SVM
↓
5 page

कि, example, types, Boosting - Random forest
काज, किछा boost काज काज,

present id: ~~26~~, ~~14~~, ~~16~~, ~~25~~, ~~12~~, ~~24~~, ~~28~~, ~~4~~, ~~10~~
~~17~~, ~~6~~, ~~27~~, ~~30~~, ~~8~~, ~~9~~, ~~5~~, ~~22~~, ~~3~~, ~~25~~, ~~10~~,
~~9~~, ~~3~~

Again class करे

Ensemble Learning: Ensemble learning:

कि, example नर definition, types ^{types are} classification
(Bagging, Random Forest, Boosting)

किछा काज करे बरन नर, example

see scenario based Qstn शुरु करे

Bagging, Boosting → working procedure.

→ 1/2 marks

SVM - মতামত, Goal, ফিচার কাক কাক।

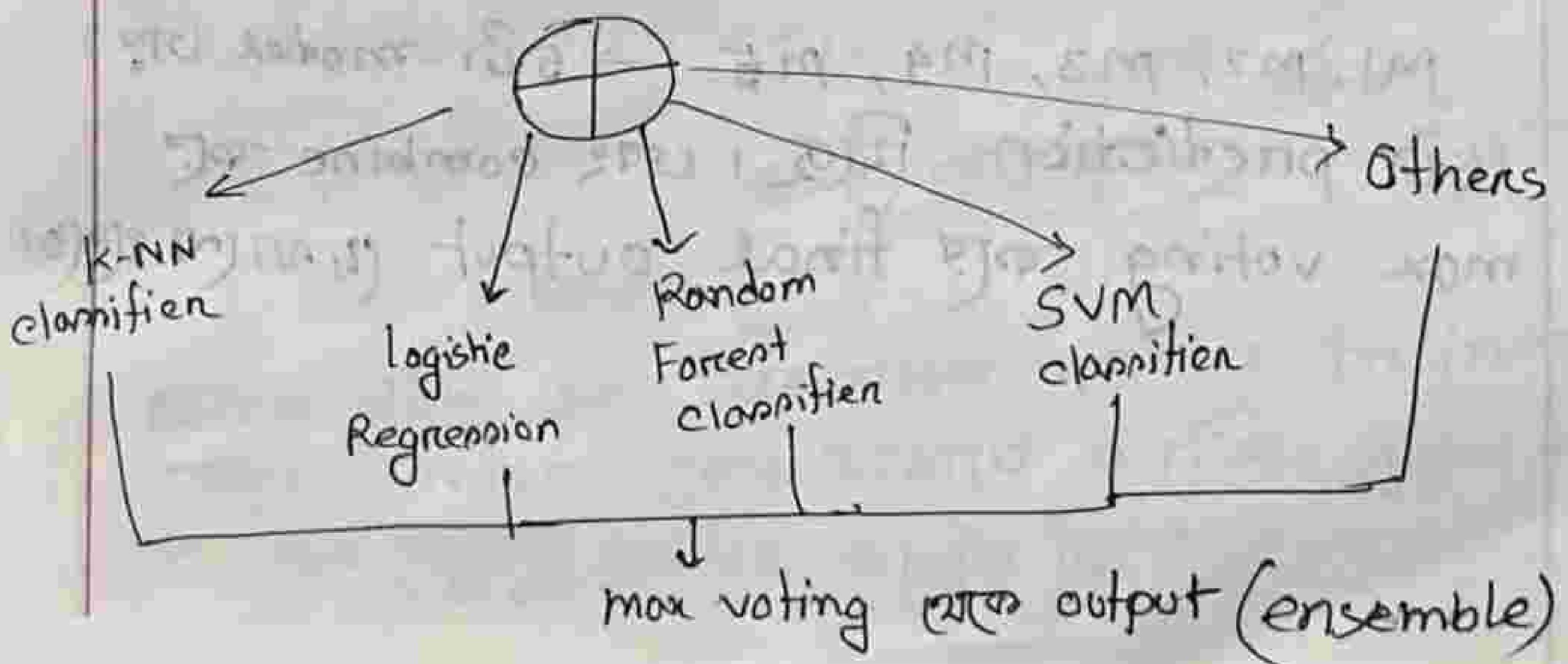
scenario based Qstn 3 মার্ক।

Youtube class

*. Ensemble Learning: it is a machine learning technique where multiple models are combined to improve overall performance.

Instead of relying on a single model, combine many methods for a better accuracy of robustness.

[it uses multiple learning algorithms to obtain better predictive performance.]



→ Study more

* Ensemble Methods (techniques) in ML:

1. Bagging or Bootstrap Aggregation

→ Bagged Decision Trees

→ Random Forest Classification or Regression

→ Extra Trees
(Random Forest)

2. Boosting

→ AdaBoost

→ Stochastic Gradient Boosting

→ XgBoost

3. Voting

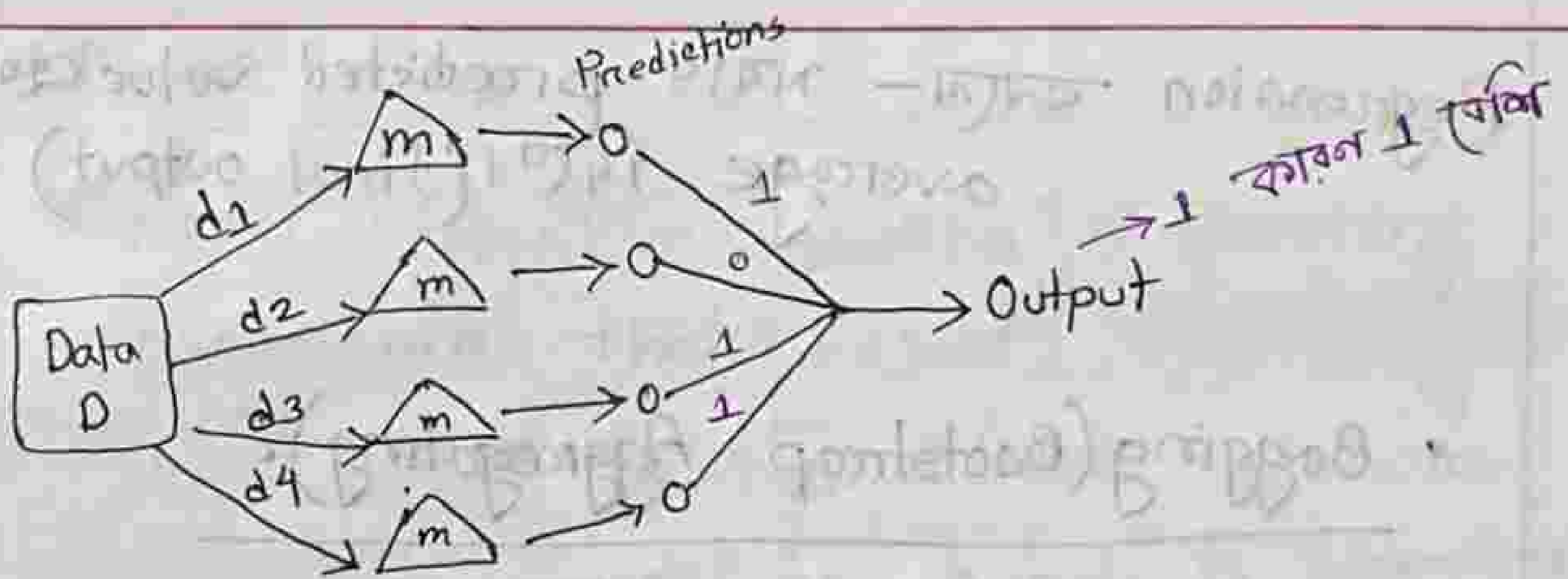
* in given slide example:

$M_1, M_2, M_3, M_4, M_5 \rightarrow$ 5 টি model যা

5 টি prediction দিচ্ছে। এবং combine করে

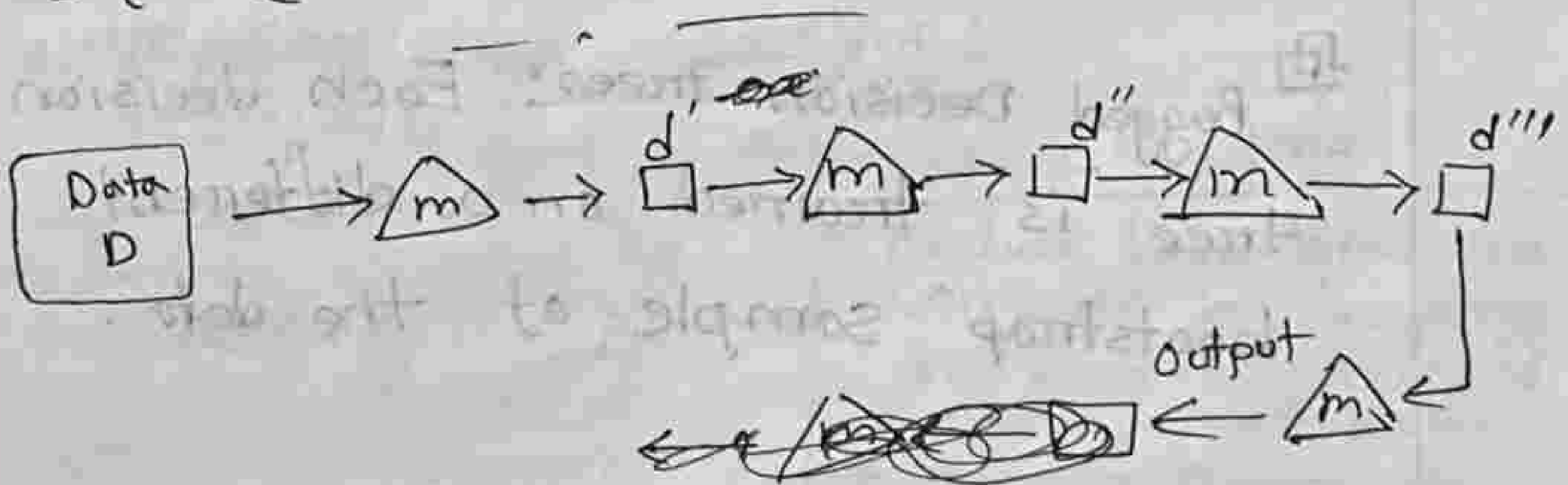
max voting করে final output দিচ্ছে।





Bagging

Data set কে ভিন্ন ভিন্ন ভাগে-
 ভাগ করে বিভিন্ন model কে train করে
 predictions value থেকে সর্বোচ্চ যে predicted
 value পাঠে (যাটা ফাইনাল ফাইনাল output
 (Maximum 1 ভাগে ফাইনাল output হিসাবে
 1 কে নেওয়া হয়েছে)।



Boosting

Same data set দিয়ে একবার-একবার train করে
 করে error rate কমাতে থাকি। যেসব data
 miss মাঝে সেসবের power বাড়িয়ে দিজে।

Regression বনাম - নব্বা predicted value এর average দিবে। (final output)

* Bagging (Bootstrap Aggregating):

☐ Random Forest: An ensemble of decision trees ^{where each tree} is trained on a bootstrap sample of the data, and the final prediction is made by averaging the predictions of all trees (for regression) or majority voting (for classification).

☐ Bagged Decision trees: Each decision tree is trained on a different bootstrap sample of the data.

* Boosting:

▣ AdaBoost (Adaptive Boosting): Boosting algorithm that adjusts the weights of incorrectly classified instances, focusing more on hard-to-classify examples in subsequent models.

▣ Gradient Boosting Machines (GBM): Algm like XGBoost, LightGBM, and CatBoost use gradient descent to minimize a specified loss function.

* Bagging Vs. Boosting:

Bagging	Boosting
→ Reduce variance by averaging multiple models	→ Reduce both bias and variance by focusing on hard-to-predict examples
→ Training Approach: parallel	→ Training Approach: Sequential
→ Equal weighting for all samples.	→ Adjusts weights: Increases weights for misclassified samples.

→ Simple averaging
(regression) or voting
(classification)

→ Improves stability
& reduces overfitting

→ Weighted sum of
model predictions

→ Improves accuracy
by focusing on
difficult cases.

x. Advantages of Ensemble Learning:

→ Improved Accuracy: By combining multiple models, ensemble methods can achieve higher predictive performance than any individual model.

→ Robustness: Ensemble methods are generally more robust to overfitting, especially if the individual models are diverse.

→ Reduction of Variance and Bias: Techniques like Bagging reduction

reduce variance, while boosting can reduce both bias & variance.

* Disadvantages of Ensemble Learning:

→ Complexity: Ensembles can be more complex to train and interpret than single models.

→ Computational cost: Can be expensive & time consuming

→ Interpretability: Making models harder to understand the overall decision making process.

Gate Smashers

↑
* Random forest: Random forests or random decision forests is an ensemble learning method for classification, regression.

Working Process:

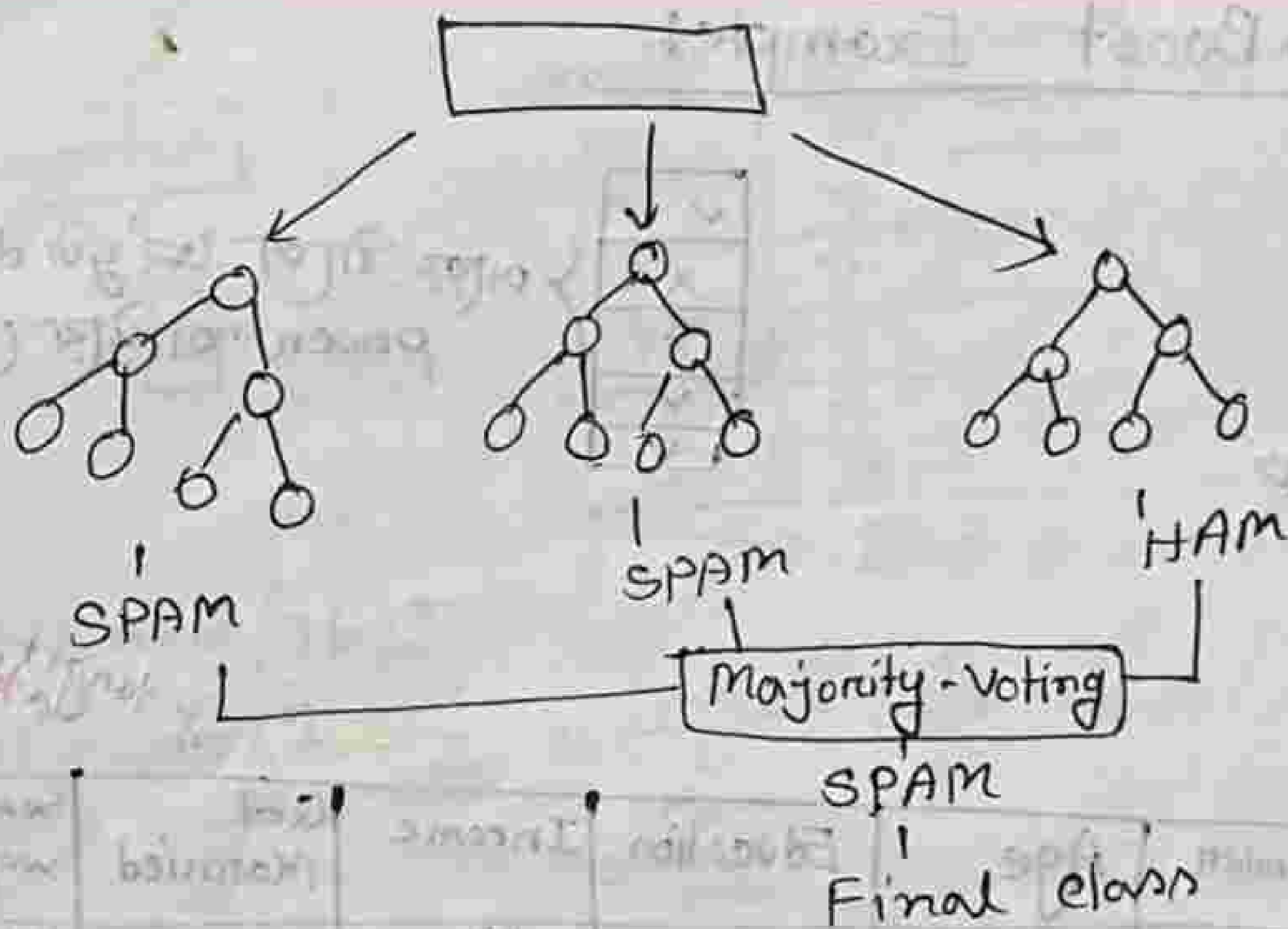
step 1: Create Bootstrap Dataset

from Original data by Randomly choosing data (repetition is allowed)

step 2: Create Randomized Decision tree from Bootstrap Dataset.

step 3: Finally output of the Random forest is the class selected by most trees (for classification) / ~~A~~ Output is average (for Regression)

ex:



Data set থেকে randomly choose করে করে
 অনেকগুলো subdataset বানাতে এবং decision tree
 দিয়ে train করাতে, অর্থাৎ output থেকে
 দেখে max vote / Average কে final output
 দিবে।

- (a) Data set
- (b) Data set
- (c) Data set
- (d) Data set

এখানে Data set

* AdaBoost Example:

✓
X
X
✓
✓

ਮਾਕਾ ਵਾਲਾ ਖਰੇ ਦੂਰੇ data ਪਾ
power ਵਾਧਿਆ (ਦਿੱਤਾ)

University
Warrriors

ex:

target
attribute

Gender	Age	Education	Income	Get Married	Sample weight
Male N	22 N	Bachelor	10 N	No	$1/n = \frac{1}{6}$
Female Y	30 Y	Masters	28 Y	Yes	$1/6$
Female Y	29 Y	Masters	31 Y	Yes	$1/6$
Male N	30 Y	Masters	37 Y	Yes	$1/6$
Male N	27 Y	Bachelor	20 N	No	$1/6$
Female Y	23 N	Bachelor	25 Y	No	$1/6$

Let,

Male = No (0)

Female = Yes (1)

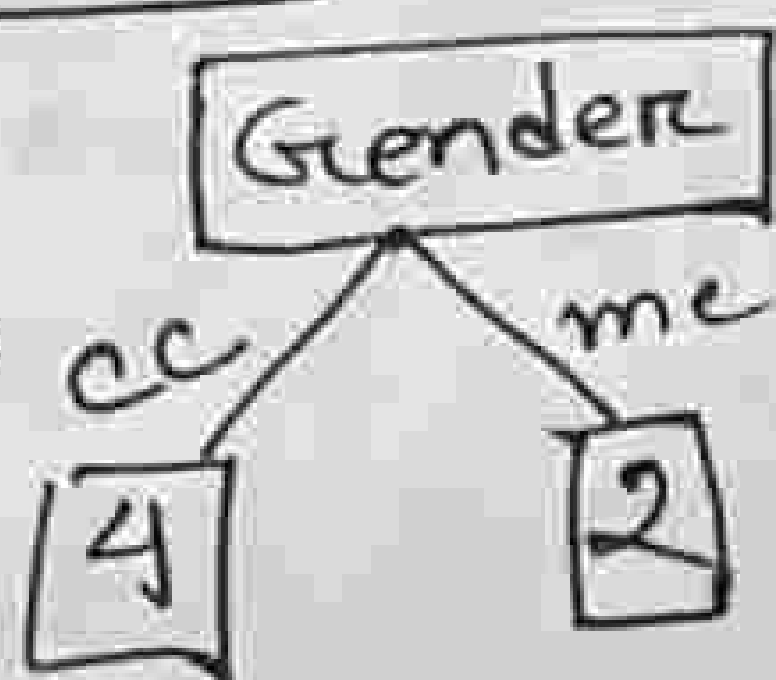
Bachelor = No (0)

Masters = Yes (1)

Qstn 2 ਨਾ ਥਾਕਣ ਵਾਲੇ ਨਿਯਮ,

Solⁿ:

Correctly
classified



Gender	Get married
No	N
Y	Y
Y	Y
N	Y
N	N
Y	N

mismatch = 2
misconnect.

Total Error (TE):

$$= \frac{2 \rightarrow mc}{2+4} = \frac{2}{6} = \frac{1}{3}$$

↓
total data

Performance say: $\frac{1}{2} \log_e \left(\frac{1-TE}{TE} \right)$

$$\Rightarrow \frac{1}{2} \ln \left(\frac{1-TE}{TE} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1-1/3}{1/3} \right)$$

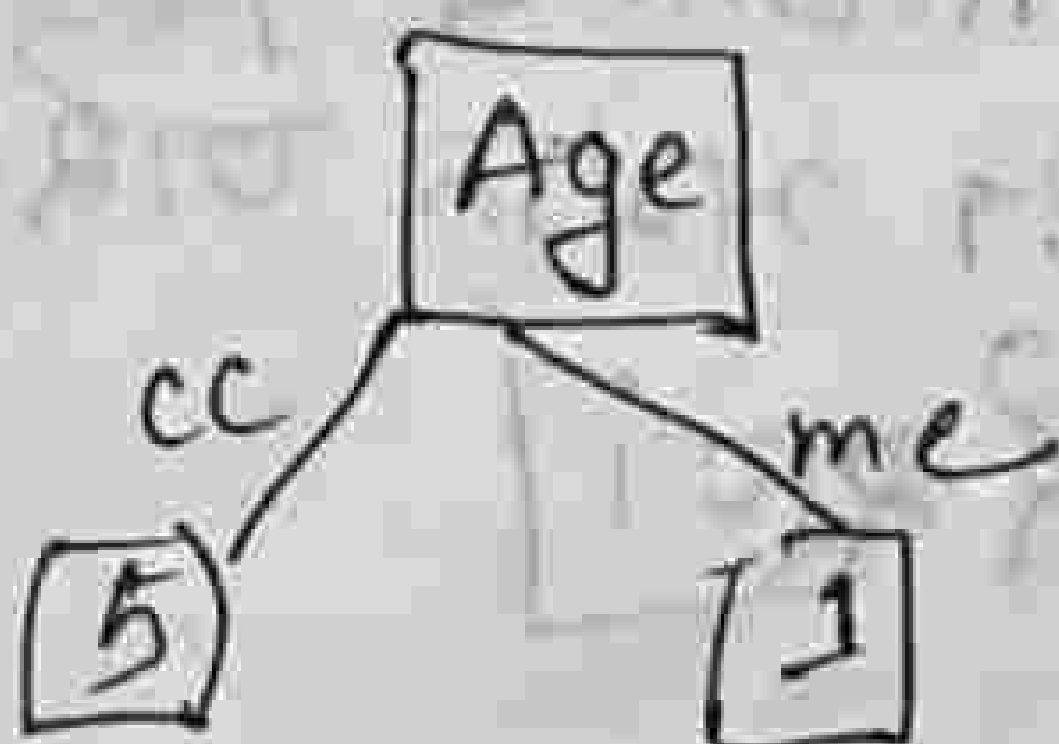
$$= 0.34$$

Again:

Age

$$Avg. = \frac{22+30+20+30+27+23}{6} = 26.66 \approx 27$$

Age = numeric value
তাই তার Average
বাব লুগুং নিচে
No এবং ঠোকে
থাকবে।



Age	Get married
N	N
Y	Y
Y	Y
N	N
Y	N

→ ০.৬৭

$$TE = \frac{1}{6}$$

Performance say:

$$\frac{1}{2} \ln \left(\frac{1-TE}{TE} \right)$$

$$= 0.804$$

Again, For education,

$$TE = 0$$

Performance say = ∞

[এটা বাদ দিও]

For Income,

$$Avg = 22.16 \approx 23$$

$$TE = \frac{1}{6}$$

Performance say = 0.804

~~Target~~ Target Attribute বা ৭ টি attribute
এর মধ্যে Age এবং Income এর performance
যদিও তাই মোটেই দুটো attribute তাই যেহেতু
একটা নিয়ম কমান্ড হবে।

Let, Income has higher performance say-

New weight for miss classify = ~~sample~~ $\text{sample weight} * e^{\text{Performance(max)}}$

$$= 1/6 * e^{0.804}$$

→ ~~2/6~~ sample weight

$$= 0.37$$

New weight for classify (correctly) :

$$= \text{sample weight} * e^{-\text{Performance(max)}}$$

$$= 1/6 * e^{-0.804}$$

$$= 0.07$$

So updated Weight:

Sample weight Income	Get married Income	sample weight	updated weight
N	N	1/6	0.07
Y	Y	1/6	0.07
Y	Y	1/6	0.07
Y	Y	1/6	0.07
Y	Y	1/6	0.07
N	N	1/6	0.07
Y	N	1/6	0.37

Normalized weight = $0.07 + 0.07 + 0.07 + 0.07 + 0.07 + 0.37$
 $= 0.72$

For cc , $\frac{0.07}{0.72} = 0.09$

For mc , $\frac{0.37}{0.72} = 0.51$

Updated w	Normalized w	clonp Range
0.07	0.09	0 - 0.09
0.07	0.09	0.09 - 0.18
0.07	0.09	0.18 - 0.27
0.07	0.09	0.27 - 0.36
0.07	0.09	0.36 - 0.45
0.37	0.51	0.45 - 0.76

আমরা
মাত্র
চারটি
চোনে

→ 9%

→ 9%

9%

9%

9%

51%

সংখ্যা ১ কিন্তু এও
চোনে শব্দ নেই।

কিন্তু সংখ্যা ৬ একটু বড়
হেঁচকি বানিয়ে নিলে প্রথমে
জামে মান কমায়।

maheeh Huddar

* Ex - From gir's slide:

Consider a training dataset of six data instances as shown in Table. Use 4 Decision Stumps for each of the 4 attributes. Apply AdaBoost algorithm & classify the dataset with Job Offer as target attribute.

CGPA	Interactiveness	Practical knowledge	Communication skill	Job Profile
≥ 9	Yes	Good	Good	Yes
< 9	No	Good	Moderate	Yes
≥ 9	No	Average	Moderate	No
< 9	No	Average	Good	No
≥ 9	Yes	Good	Moderate	Yes
≥ 9	Yes	Good	Moderate	Yes

Soln.

Step 1:

Initial weight assigned to each item
 $= 1/6$

Step 2: Iterate for each weak classifier

Decision Stump For CGPA,

যদি $CGPA \geq 9$ হলে

if $CGPA \geq 9$,
the data instance
is predicted to have
Job offer as Yes else No

CGPA	Predicted Job offer	Actual Job offer	Weight
≥ 9	Y ✓	Y	1/6
< 9	N ✗	Y	1/6
≥ 9	Y ✗	N	1/6
< 9	N ✓	N	1/6
≥ 9	Y ✓	Y	1/6
≥ 9	Y ✓	Y	1/6

Weighted error: $E_{CGPA} = \frac{1/6 + 1/6}{6}$ or $2 * \frac{1}{6}$
 $= 0.333333$

Weak classifier:

$$\alpha_{CGPA} = \frac{1}{2} \times \ln \frac{(1 - E_{CGPA})}{E_{CGPA}}$$

$$= 0.34657$$

Normalizing Factor:

$$Z_{CGPA} = \text{wt}(\text{correct classified instance}) \times \text{No. of correct classification} \times e^{-\alpha_{CGPA}} + \text{No. of wt (wrong classified ins.)} \times \text{No. of wrong classification} \times e^{+\alpha_{CGPA}}$$

$$= \frac{1}{6} \times 4 \times e^{-0.347} + \frac{1}{6} \times 2 \times e^{0.347}$$

$$= 0.9428$$

or update weight এর মান
যদি ভুল ক্লাস
যাওয়া যায়
previous math
দেখানো শ্র(৬)

Update weight :

$$w_{\text{new}} = \frac{w_{\text{old}} \times e^{-\alpha}}{Z} \quad \left[\text{for correctly classified} \right]$$

normalization factor

$$= \frac{\frac{1}{6} \times e^{-0.347}}{0.9428} = 0.1249$$

$$w_{\text{new}} = \frac{w_{\text{old}} \times e^{+\alpha}}{Z} \quad \left[\text{For miss correctly classified data} \right]$$

$$= \frac{\frac{1}{6} \times e^{0.347}}{0.9428} = 0.2501$$

Now,

CGPA	Predicted Job offer	Actual Job offer	Weight
≥ 9	Y	Y	0.1249
< 9	N	Y	0.2501
≥ 9	Y	N	0.2501
< 9	N	N	0.1249
≥ 9	Y	Y	0.1249
≥ 9	Y	Y	0.1249

যা হোক misclassified হওয়ায় তাহলে weight বাড়িয়ে দেওয়া হল। $\rightarrow$ আল্ট্রাস্ট্রাস্ট

Decision Stump for Interactiveness:

	Interactiveness	Predicted Job offer	Actual Job offer	Weight
2	Y	Y	Y	0.1249
2	N	N	Y	0.2501
2	N	N	N	0.2501
2	N	N	N	0.1249
6	Y	Y	Y	0.1249
8	Y	Y	Y	0.1249

Weighted Error, $\epsilon_{\text{Int.}} = 1 \times 0.2501$
 $= 0.2501$

Weak classifier,

$$\alpha = \frac{1}{2} \ln \frac{(1 - \epsilon_{\text{Int}})}{\epsilon_{\text{Int}}}$$

$$= \frac{1}{2} \times \ln \left(\frac{1 - 0.2501}{0.2501} \right)$$

$$= 0.5490$$

Normalization Factor,

$$\bar{z} = 0.1249 \times 5 + 0.2501 \times 1$$

$$\bar{z} = 0.1249 \times 4 + 0.2501 \times 1$$

$$\bar{z} = [0.1249 \times 4 \times e^{-0.5490}] + [0.2501 \times 1 \times e^{0.5490}]$$

$$= 0.866$$

Start factor,

if, yes $\rightarrow$ Yes
 No $\rightarrow$ No

Update weight:

$$W_{\text{new}} = \frac{W + e^{-\alpha}}{2} \quad [\text{for } \text{cc}]$$

$$= \frac{0.2501 \times e^{-0.5490}}{2}$$

$$= 0.0832$$

$$= 10.0832$$

$$W_{\text{new}} = \frac{W + e^{\alpha}}{2} = \frac{0.1249 \times e^{0.5490}}{0.866}$$

$$= 0.5001 = \cancel{0.1667} [\text{for } \text{mc}]$$

$$W_{\text{new}} = \frac{W + e^{-\alpha}}{2} = \frac{0.2501 \times e^{-0.549}}{0.866} = \cancel{0.5001} = 1.6667$$

Row = 3

weight diff.

Now,

Interative	Predicted Job offer	Actual Job offer	weight
Y	Y	Y	0.0832
N	N	Y	0.5001
N	N	N	0.1667
N	N	N	0.0832
Y	Y	Y	0.0832
Y	Y	Y	0.0832

પ્રજ્ઞા સમજાવેલું (વર્ગ કક્ષણ, slide લઈ મળે,

મતી કામ 41 page slide -

Sl. No. 1:

$$\alpha_{\text{control}} \times Y + \alpha_{\text{interact}} \times Y_{\text{en}} + \alpha_{\text{csi}} \times Y_{\text{en}}$$

$$= 0.347 \times 1 + 0.5490 \times 1 + (-0.5485) \times 1 = 0.3475$$

↓

Final prediction

Yes કાળ

0 (અલગ)

Sl. No. 2:

$$0.347 \times 0 + 0.5490 \times 1 - 0.5485 \times 0 = 0.5490$$

↓
No

Yes શબ્દ 1 દ્વારા મૂલ્ય
No શબ્દ 0 દ્વારા મૂલ્ય

* Soft Margin Vs Hard Margin:

Soft

→ Used when data is not linearly separable in feature space

→ Allows data points inside the margin

→ Soft margin requires cost parameter

Hard

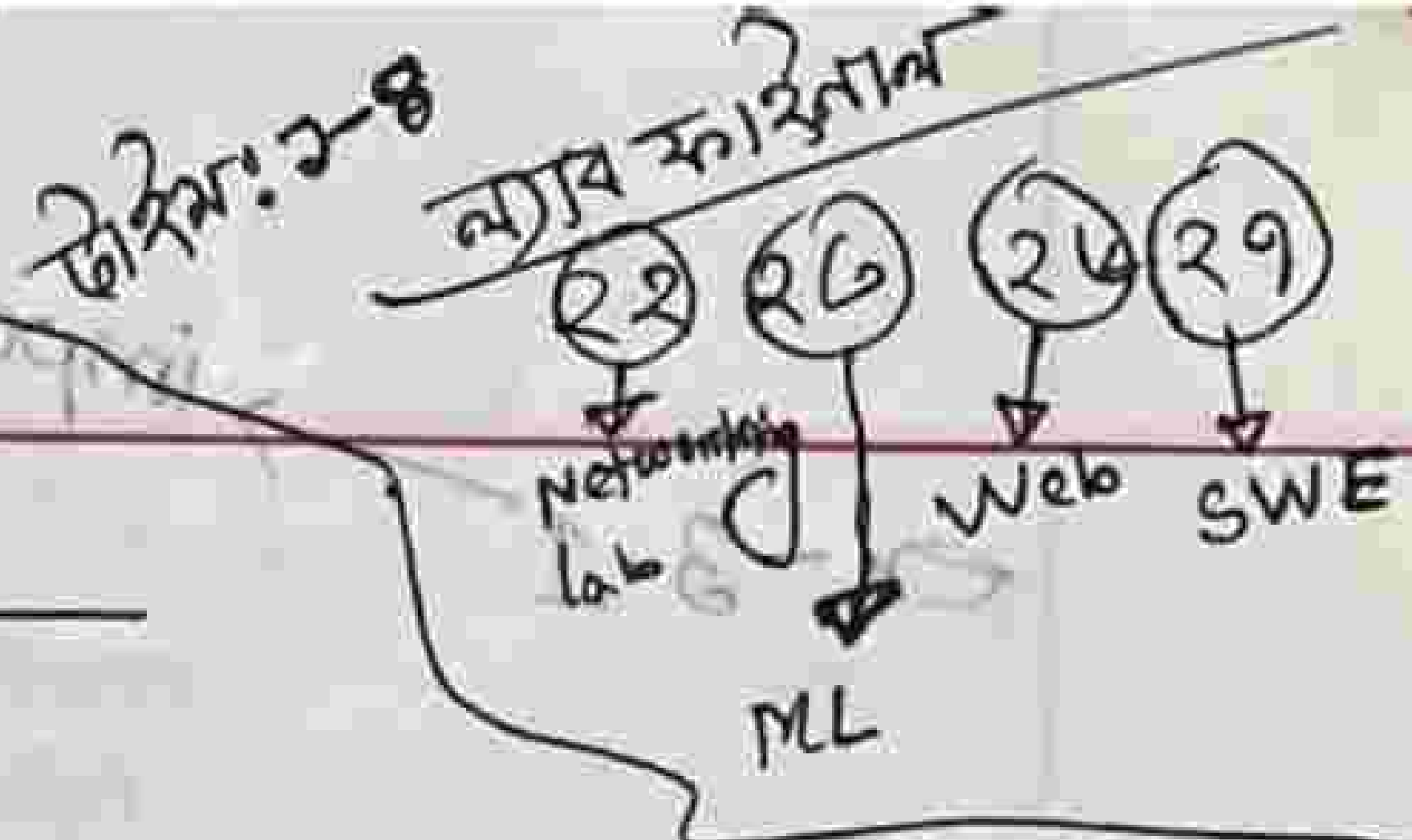
→ Used when data is linearly separable in feature space.

→ doesn't allow data points inside the margin

→ doesn't require

short-form
↑
FLD
↑
short-form

20-04-2026



* Fisher's Linear Discriminant Analysis :

3.2 Fisher's Criterion: ২ বা তার বেশি সংখ্যক ডাটা'র projection করা way.

slide ৬
সুন্দর করে প্রমাণ করুন
যে $d = \text{distance}$

data set এর mean value এর
এবং mean value থেকে দূরত্ব
কর then classification

FLD Maximizes the following Ratio :

$$J(w) = \frac{\text{Between-class variance}}{\text{Within-class variance}}$$

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

S_B = Between class scatter matrix

S_W = within " " "

w = projection vector.

* Importance & And Advantages of FLD

↪ দ্রুত করা
সহজ

৫.২: → এই পার্সন

৫.২: → math given equation related

$$z = wx + b$$

Lecture sheet 2:

PDM → page 1

Probabilistic Dis. Model

Basic idea, equation(x)

why imp., advantage & disad

page - 3 পার্সন

২ম অংশ

ML:



Theory + ~~math থাকবে~~ math
+ math থাকবে

slide এর topics (যেকোনো গুটন হইছে)

(equation থেকে math) → sob slide (যাক)

→ Accuracy এর কথা **

→ F_1 , recall, Precision

→ data / hints থেকে

math

performance evaluation → formula

bias, variance trade-off:

Graph বুঝতে হবে,

figure মনে

Ehsan
Mahmud

Linear Regression: দুই types (simple, multiple, polynomial, Ridge, Lasso)

Theoretical, কোন type কি জিনিষ,
২টা/৩টা মতে difference

Types - সুন্দর Criteria, কোনটা কোন ক্রমে
করা।

Logistic Regression:

Cost Function

Sigmoid Func. $\rightarrow$ *

Cost Func, Cost Funct. expand
করা techniques

Formula: * update করা কিভাবে, আনফলুয়া
step.

$\rightarrow$ matrix থেকে y এর করে
update করা.

DT:

কি, কোন use
করি

Entropy বিয়. করা,

Entropy $\rightarrow$ full theory + math

* DT $\rightarrow$ ২০০০ দিকে বাংলা - উইল,

33 page - I.G. প্রভৃতি

Ensemble learning: Types, Bagging-Boosting

6 page $\rightarrow$ classification

Bagging-Boosting $\rightarrow$ कि, कस, किछाए

11 page Random Forest: कि,
किछाए काङ्कए, advantage

24 page $\rightarrow$ ex + advantage

26 page $\rightarrow$ math